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Graduation Project

To obtain the
Engineering Degree

in Applied Sciences and Technology

Specialty: Computer Networks and Telecommunications

Intelligent Distributed Network Softwarization to Alleviate Internet of Things Constraints

Presented by

Mohamed Ali Zormati

Hosted by Heudiasyc UMR UTC/CNRS 7253



Defended on September 19, 2023 in front of a jury composed of

Dr. Bassem BEN SALAH President of the Jury

Dr. Meriem AFIF Reviewer

Dr. Hicham LAKHLEF Host Institution Supervisor

Dr. Sofiane OUNI INSAT Supervisor

Academic Year: 2022/2023



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Host Institution Supervisor Dr. Hicham LAKHLEF	INSAT Supervisor Dr. Sofiane OUNI
September 7, 2023	September 7, 2023

Academic Year: 2022/2023

*Great things are done by a series
of small things brought together.*

VINCENT VAN GOGH

Dedication

I dedicate this work to my beloved family and dear friends, whose unwavering support and encouragement have been the pillar of my academic journey. Your belief in me has been a constant source of inspiration, and I am deeply grateful for your presence in my life. This accomplishment is as much yours as it is mine, and I share it with all of you with heartfelt appreciation.

With gratitude,

Mohamed Ali Zormati

September 7, 2023

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This work would not have been possible without valuable contributions and unwavering dedication. Through this manuscript, I would like to pay tribute to all those who have contributed to this work in one way or another.

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My deepest gratitude goes to the entire academic and administrative team of the National Institute of Applied Sciences and Technology (INSAT) for their guidance and support over the past five years, both in my academic and volunteer experience. This has been extremely beneficial in enhancing my skills.

Finally, I would like to express my sincere appreciation to the honorable members of the jury for taking the time to evaluate this work and I hope that its quality meets their rigorous expectations.

Abstract

The Internet of Things (IoT) has grown significantly, evolving from a novel technology to an integral part of everyday life. It connects a massive number of devices, which collect valuable data through various sensors, to each other and to the Internet. The tremendous volume of interconnected heterogeneous and constrained devices poses enormous challenges as IoT provides diverse and complex network services with different requirements on a common infrastructure.

Network softwarization is an accurate solution to these concerns, as it has emerged as a paradigm that enhances traditional networking by decoupling hardware from software through the use of key technologies such as Software Defined Networking (SDN) and Network Function Virtualization (NFV). Machine Learning (ML) plays a pivotal role in building efficient and intelligent networks as it has shown impressive results in various domains. As network softwarization makes it easy to integrate intelligence, the combination of SDN, NFV, and ML is key to building efficient and agile IoT networks.

In this work, after surveying the state of the art, we propose an enabling architecture for intelligent distributed network softwarization for the Internet of Things, taking advantage of the combination of IoT, SDN, NFV, and ML. We also propose a novel routing mechanism for softwarized IoT networks based on Federated Deep Reinforcement Learning (FDRL). We finally conduct experiments to establish the effectiveness of our proposal, which outperforms its conventional counterparts.

Keywords: Internet of Things (IoT), Network Softwarization, Software Defined Networking (SDN), Network Function Virtualization (NFV), Machine Learning (ML).

Résumé

L’Internet des objets (IoT) est en pleine expansion, passant d’une technologie nouvelle à une partie intégrante de la vie quotidienne. Il connecte un nombre considérable d’objets, recueillant des données utiles à travers différents capteurs, entre eux et l’Internet. Le volume énorme d’objets interconnectés, hétérogènes et avec des ressources limitées, pose d’énormes défis, puisque l’IoT a à fournir une variété de services réseau complexes avec des exigences différentes, le tout sur une infrastructure commune.

La softwarisation de réseau constitue une solution intéressante à ces problèmes. Elle est apparue comme un paradigme qui améliore les réseaux traditionnels en dissociant le matériel du logiciel grâce à l’utilisation de technologies clés telles que les réseaux définis par logiciel (SDN) et la virtualisation des fonctions réseau (NFV). L’apprentissage automatique (ML) joue un rôle clé dans la création de réseaux efficaces et intelligents, avec des résultats impressionnants dans différents domaines. Puisque la softwarisation de réseau facilite l’intégration de l’intelligence, la combinaison de SDN, NFV et ML est essentielle pour créer des réseaux IoT efficaces et agiles.

Dans ce travail, après avoir fait le point sur l’état de l’art, nous proposons une nouvelle architecture pour la softwarisation intelligente et distribuée des réseaux IoT, combinant IoT, SDN, NFV et ML. Nous proposons également un mécanisme de routage novateur pour les réseaux IoT softwarisés basé sur l’apprentissage par renforcement profond fédéré (FDRL). Enfin, nous réalisons des expériences pour déterminer l’efficacité de notre proposition, qui dépasse ses contreparties conventionnelles.

Mots-clés : Internet des Objets (IoT), Softwarisation de Réseau, Réseaux Définis par Logiciel (SDN), Virtualisation des Fonctions Réseau (NFV), Apprentissage Automatique (ML).

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List of Acronyms

- **AI:** Artificial Intelligence
- **ANN:** Artificial Neural Network
- **API:** Application Programming Interface
- **BMU:** Best Matching Unit
- **CAPEX:** CApital EXpenditures
- **CFL:** Centralized Federated Learning
- **CNN:** Convolutional Neural Network
- **DAG:** Directed Acyclic Graph
- **DDoS:** Distributed Denial of Service
- **DFL:** Decentralized Federated Learning
- **DL:** Deep Learning
- **DNN:** Deep Neural Network
- **DRL:** Deep Reinforcement Learning
- **DRL-R:** Deep Reinforcement Learning - Routing
- **EC:** Edge Computing
- **FDRL:** Federated Deep Reinforcement Learning
- **FL:** Federated Learning
- **FRL:** Federated Reinforcement Learning
- **FTL:** Federated Transfer Learning
- **GMM:** Gaussian Mixture Model
- **GNN:** Graph Neural Network

- **HFL:** Horizontal Federated Learning
- **IDE:** Integrated Development Environment
- **IEEE:** Institute of Electrical and Electronics Engineers
- **IoT:** Internet of Things
- **k -NN:** k -Nearest Neighbor
- **LSOA:** Lion Swarm Optimization Algorithm
- **MADQN:** Multi Agent Deep Q-Network
- **MANO:** Management ANd Orchestration
- **MDP:** Markov Decision Process
- **MDS:** Markov Decision Support
- **MitM:** Man in the Middle
- **ML:** Machine Learning
- **MLP:** MultiLayer Perceptron
- **NFV:** Network Function Virtualization
- **NFVI:** Network Function Virtualization Infrastructure
- **OPEX:** OPerational EXpenditures
- **P2P:** Peer to Peer
- **PDR:** Packet Delivery Ratio
- **QoE:** Quality of Experience
- **QoS:** Quality of Service
- **RF:** Random Forest
- **RL:** Reinforcement Learning
- **RNN:** Recurrent Neural Network
- **RPL:** Routing Protocol for Low-power and lossy networks
- **RTT:** Round-Trip Time
- **SDN:** Software Defined Networking
- **SFC:** Service Funtion Chain
- **SL:** Supervised Learning

- **SOM:** Self-Organizing Map
- **SPoF:** Single Point of Failure
- **SPR:** Shortest Path Routing
- **UL:** Unsupervised Learning
- **VFL:** Vertical Federated Learning
- **VNF:** Virtualized Network Function
- **VNF-FG:** Virtualized Network Function - Forwarding Graph

List of Publications

International conferences

- **Zormati M.A.**, Lakhlef H.: "An Overview of Machine Learning-Enabled Network Softwarization for the Internet of Things", 31st International Conference on Software, Telecommunications and Computer Networks (SoftCOM), Split, Croatia, 2023.
- **Zormati M.A.**, Lakhlef H.: "Enabling Architecture for Distributed Intelligent Network Softwarization for the Internet of Things", 20th IEEE International Conference on Mobile Ad Hoc and Smart Systems (MASS), Toronto, Canada, 2023.
- **Zormati M.A.**, Lakhlef H.: "Routing Optimization Based on Distributed Intelligent Network Softwarization for the Internet of Things", 39th ACM SIGAPP Symposium on Applied Computing (SAC), Avila, Spain, 2024 (**submitted**).

International journals

- **Zormati M.A.**, Lakhlef H.: "Review and Analysis of Recent Advances in Intelligent Network Softwarization for the Internet of Things", Computer Networks, Elsevier, 2023 (**submitted**).

General Introduction

The Internet of Things (IoT) is a promising technology that establishes connectivity of anything from anywhere at anytime [1]. Today, the IoT landscape includes billions of heterogeneous devices that affect every aspect of human life. It is growing exponentially, with the number of devices expected to exceed 20 billion by 2025 [2]. The huge number of interconnected devices poses new challenges to IoT networks as these devices generate a massive and increasing amount of data [3]. Managing such a heterogeneous network poses technical challenges, and to overcome these obstacles, IoT networks should be combined with other technologies, such as network softwarization [4].

Network softwarization is a novel key technology for building dynamic and agile IoT networks. It encompasses both Software Defined Networking (SDN) and Network Function Virtualization (NFV) and aims to transform the communication process and system components from legacy network devices, where software is tightly bundled with hardware [5], to general-purpose devices that enable a wide range of services while ensuring remarkably low CApital and OPerational EXpenditures (i.e., CAPEX and OPEX) [4]. However, many challenges (e.g., security concerns, resource utilization) must be fully addressed for effective network softwarization [6].

Machine Learning (ML), a category of Artificial Intelligence (AI), is a promising approach to bring intelligence to the network [7] and can be used to address these issues and enable effective network softwarization for IoT. ML techniques enable networks to learn from their experience, increasing their resilience to vulnerabilities and failures while optimizing performance.

Although many research efforts have been made to investigate these three promising technologies and their combination, to the best of our knowledge, few work has actually focused on intelligent (i.e., ML-enabled) network softwarization for IoT networks.

Therefore, our work aims to fill this gap. After reviewing the state-of-the-art solutions and identifying the most prominent future directions, we propose a novel intelligent distributed network softwarization architecture for IoT. It combines SDN, NFV, and ML with IoT to increase the overall performance while addressing the challenges. We use this general architecture as a basis to propose a novel intelligent routing mechanism for softwarized IoT based on Federated Deep Reinforcement Learning (FDRL), which is applied to tackle routing challenges in such a heterogeneous networking environment. We proceed with simulation to evaluate the effectiveness of our contribution.

The remainder of this report is organized as follows. First, in Chapter 1, we present the project framework by introducing the host institution, the project context, and the adopted work methodology and timeline. In Chapter 2, we provide a background on IoT, network softwarization with a focus on SDN and NFV, and ML. Chapter 3 presents a comprehensive overview of existing intelligent network softwarization for IoT, followed by a discussion and identification of future directions. Then, in Chapter 4, we present our contribution to ensure intelligent distributed network softwarization for IoT, and its application in routing optimization. We perform simulations and performance evaluations to prove the effectiveness of our proposal, in Chapter 5. Finally, we conclude this work and present the main future directions.

1

Project Framework

Introduction

In this first chapter, we present the framework of the project. First, we introduce the host institution. Then we present the project context and motivation behind. Finally, we present the work methodology and the adopted timeline.

1 Host institution

This work is carried out at the Heudiasyc Laboratory of the University of Technology of Compiègne, within the framework of the Safety, Communication and Optimization (SCOP) team.

1.1 University of Technology of Compiegne

The University of Technology of Compiegne (UTC) is a scientific, cultural, and professional public establishment founded in 1972 by the French government as its first experimental university of technology, located in Compiegne, Oise, France. Whether in terms of education, research, or industrial links, UTC is regularly ranked among the top engineering schools [8]. It is part of the Sorbonne University Alliance.



Figure 1.1: UTC Logo

According to its official website [8], its main assigned missions are: to ensure excellent thematic research, open to the issues of its environment, to produce knowledge, skills, and expertise, to transmit and decimate knowledge through teaching and training, and to introduce knowledge in a transformation process that fosters the production of inventions.

Eight research units form the core of the UTC's research excellence, many of which are affiliated with the French National Center for Scientific Research (CNRS), of which the Heudiasyc laboratory is one.

1.2 Heudiasyc laboratory

The Heudiasyc laboratory (that is, heuristics and diagnostics for complex systems) was created in 1981 and has the code *UMR UTC/CNRS 7253* as a joint research unit. Since its creation, it has been closely associated with the CNRS, one of the world's leading research organizations [9]. Heudiasyc research is in the field of information and communication sciences, namely computer science, automatic control, robotics, and artificial intelligence [10]. According to the 2022 annual report [11], the team is composed of around 130 researchers, complemented by around 25 trainees per year. At a glance, Heudiasyc has 40 ongoing projects, 50 ongoing PhD theses, and a publication rate of about 160 international publications per year. It is collaborating with more than 30 industrial partners and 10 international organisms.

Heudiasyc is composed of three research teams: Knowledge, Uncertainty, and Data (CID), Safety, Communication, and Optimization (SCOP) and Robotic Systems in Interaction (SyRI).



Figure 1.2: Heudiasyc Logo

This work is carried out within the SCOP team. SCOP research focuses on the design and optimization of logistics and networked systems, as well as safe and secure systems. SCOP's areas of interest are reliability, safety, and optimization, and the interplay between these three aspects. Areas of application are complex systems and systems of systems [10]. It is headed by Pr. Dritan Nace since January 2022.

The main scientific axes of the team in the latest years are: safe and secure systems (e.g., uncertainty management in dependability studies, fault tolerance, system and network security), communication systems (e.g., dynamic mobile networks, autonomous and energy-efficient networks), and system optimization (e.g., planning and scheduling, network optimization) [11].

2 Project context

As IoT has grown significantly in a wide variety of applications (e.g., smart homes, smart cities, smart vehicles) [12], the challenges have increased accordingly due to the large number of heterogeneous and constrained devices. This number is growing exponentially, although the growth has been slowed by the COVID-19 pandemic, as shown in Figure 1.3, courtesy of IoT Analytics [13].

As such, IoT networks face several challenges (e.g., scalability, energy efficiency, security) [1] that must be urgently addressed to support the growth rate. When combined with IoT, network softwarization addresses network concerns while providing some benefits (e.g., programmability) [4]. When intelligence is incorporated, it ensures that networks are self-aware, self-adaptive, and self-managed [6].

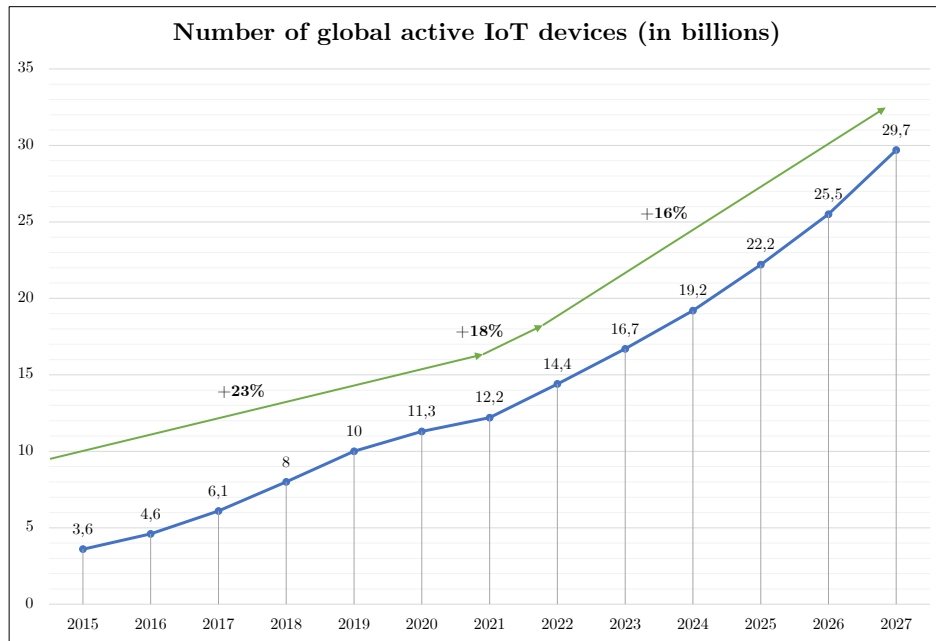


Figure 1.3: Number of Connected IoT Devices [13]

Accordingly, the main objective of this project is to study the combination of IoT, network softwarization, and ML to alleviate the constraints and increase the efficiency and scalability of IoT networks. This is motivated by the advantages provided by softwarization and the success in introducing intelligence in various domains.

3 Work methodology and timeline

As Szent-Györgyi, a Nobel Prize-winner, once said [14], "*research is to see what everybody else sees, and think what nobody else has thought*". Therefore, since the current work is a research project, we have adopted a work methodology defined as follows.

1. We first reviewed the background concepts of IoT, network softwarization with emphasis on SDN and NFV, and ML. This step helps to speed up the next steps, thus improving the work efficiency.
2. We then conducted an initial quick literature review to identify the core objective of the project, which is intelligent network softwarization. This step is critical because it defines the scope of the project in accordance with the preliminary topic of the internship.

3. While defining the cap, we surveyed the state of the art of intelligent network softwarization for IoT and identified the main research challenges and future directions. This step is crucial, as it avoids scattered research and allows to improve and build on the existing work.
4. As the contribution axes were set, we first proposed a novel distributed intelligent network softwarization architecture for IoT. This architecture is the framework for the next contributions.
5. We then proposed to address one of the most prominent challenges in IoT networks, namely routing, based on our proposed architecture. We designed the solution based on the existing work.
6. Finally, we performed simulations to evaluate the effectiveness of the solution compared to its counterparts.
7. We concluded our work by identifying the future directions based on the contributions and the faced challenges. This step lays the baseline for future work.

Accordingly, we now present the inherent timeline that was adopted to divide the tasks during the six-month internship in the laboratory, as illustrated by the Gantt chart in Figure 1.4.

ID	Task	April 2023		May 2023		June 2023		July 2023		August 2023		September 2023	
		1-15	16-30	1-15	16-31	1-15	16-30	1-15	16-31	1-15	16-31	1-15	16-30
1	Understanding the topic (background)	■											
2	Literature comprehensive review		■	■	■	■							
3	Conference survey paper submission		■	■									
4	Journal survey paper submission			■	■	■							
5	Solution architecture					■	■						
6	Architecture conference paper submission					■							
7	Preparations for the implementation						■	■					
8	Simulation environment setup						■						
9	Softwarization environment setup							■					
10	Intelligent softwarization for IoT networks							■	■				
11	Migration to a distributed solution								■	■			
12	Solution performance evaluation									■	■		
13	Work conclusion and reporting											■	■
14	Solution conference paper submission												■

Figure 1.4: Internship Timeline

We also report that a number of research papers were submitted to international conferences and accepted for presentation and publication in the proceedings, as stated earlier in the report. A review article has also been submitted for publication in a journal, but is still under review due to significant delays in the process. A conference paper presenting the proposed solution and the experimental results has been submitted to conclude this work.

Conclusion

In this chapter, we have framed the project by introducing the host institution, the context, and the methodology adopted. Now that the motive behind the work is established, we will proceed by introducing the main background notions of IoT, network softwarization, and ML techniques.

2

Background

Introduction

As a prelude to surveying the state of the art, we give an outline of the main concepts of IoT, network softwarization with an emphasis on SDN and NFV technologies, and ML techniques. IoT embodies the connection of a wide range of objects to the Internet, creating a networked environment ranging from everyday objects to complex industrial systems. Considered as a transformative approach, network softwarization involves reconfiguring and managing the network infrastructure using software-based techniques. Two major technologies within this framework are SDN, which centralizes network control and management, and NFV, which virtualizes network services to enable more flexible and efficient network provisioning. In our context, ML techniques are called upon to enhance the intelligence and adaptability of networks and thus optimize their performance and efficiency.

1 Internet of Things

The Internet of Things (IoT) brings together billions of devices (i.e., things) to provide a wide array of services by establishing connectivity between these devices and the Internet. In addition to connecting things, IoT also focuses on optimizing and automating traditional systems, as evidenced by the various IoT applications [15].

1.1 IoT applications

The global growth of IoT has made it an integral part of everyday life. Today, it is deployed in a wide variety of domains, enhancing traditional systems. Below we list some of the most recent use cases of IoT [16] [17], and a broader perspective of the applications diversity is showcased in Figure 2.1.

- **Healthcare:** Integrating IoT into medical devices can improve the quality of healthcare services. In an emergency, a quick response can be a matter of life and death, and IoT helps strengthen the connection between patients and medical staff.
- **Agriculture:** IoT can be used to control climatic conditions to maximize fruit production, monitor weather conditions, etc. It helps in automating manual processes (e.g., watering), while giving farmers visibility to optimize resources.
- **Environment monitoring:** The use of IoT enables the detection and prevention of natural disasters such as earthquakes [18], landslides, etc. Through IoT, stakeholders can collect and analyze data remotely, reducing the need for costly on-site visits.



Figure 2.1: Cutting-Edge IoT Use Cases [16]

1.2 IoT architecture

Several architectures have been proposed for IoT [19]. According to [15], most of the proposals consist of three layers, as shown in Figure 2.2.

- **Perception layer:** Contains the objects used for sensing. This layer is responsible for sensing the environment, collecting the sensing data, and sending it to the next layer (specifically, to the IoT gateway).
- **Network layer:** Contains the network devices and communication technologies. This layer is considered the interface between the perception layer and the application layer.
- **Application layer:** Contains the IoT services and applications. This is the front end of the system as it is an interface between the IoT system and the users.

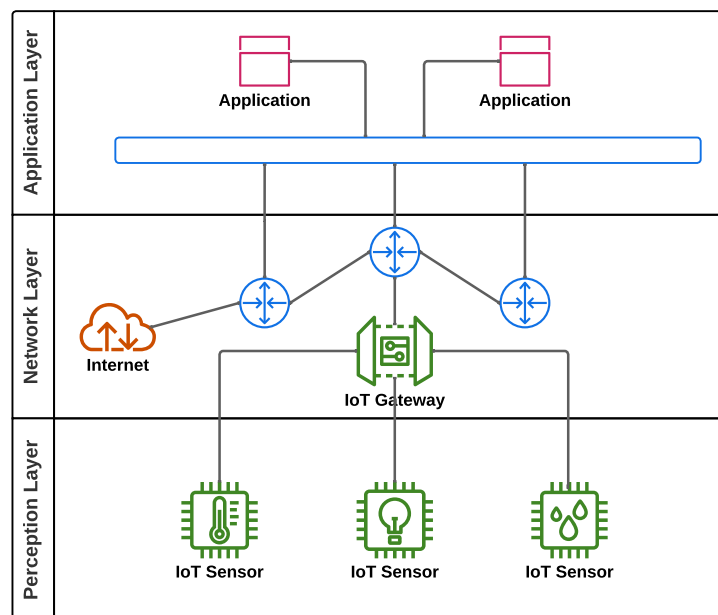


Figure 2.2: IoT Architecture

Note that some architectures add a fourth layer, called the **Middleware layer**. It coordinates between IoT devices from different vendors, allowing communication between all objects regardless of their hardware [15]. This provides more flexibility in the association between devices [19]. The middleware layer is responsible for service management across IoT devices in order to establish connections between the ones that provide the same service [20].

1.3 Major challenges

The complexity of IoT arises primarily from the vast multitude of interconnected devices accessing the Internet and interacting with one another. This complexity is further compounded by the diversity of protocols and models that govern their operations [15]. In fact, IoT faces several challenges and has specific requirements, which are detailed below [17] [19] [20] [21].

- **Heterogeneity:** It comes into picture because IoT can connect any physical object. Therefore, a unified architecture is required to implement IoT properly.
- **Scalability:** It needs to be considered while designing solutions for IoT (e.g., routing protocols and data storage mechanisms).
- **Energy efficiency:** Since most IoT devices are self-powered, the power consumption of each node must be carefully considered and optimized.
- **Software development challenges:** Consider the 5Vs of big data (volume, variety, velocity, veracity, and value) and self-configuration (software, hardware, and network configuration).
- **Standardization:** There is currently a lack of standardization, which limits the effectiveness and performance of the IoT.
- **Security challenges:** For example, IoT systems must consider the necessary criteria for establishing computer system security (e.g., confidentiality, integrity, etc.).

To address the above issues and challenges, and to meet the quality requirements of IoT, many research works propose network softwarization as a key enabling technology for IoT networks.

2 Network softwarization

Network softwarization is an emerging approach [22] to transform the traditional networks to the new trending technologies such as programmable network and virtualization. By leveraging the software features such as flexibility, it aims to provide new functionalities and deploy them efficiently [15].

The goal is to transform the network into an open ecosystem that decouples hardware and software components. This transformation is expected to deliver benefits such as advanced services, enhanced networking capabilities, and improved network development and maintenance capabilities [23]. The following features show the importance of network softwarization: separation of responsibilities (by separating data and control planes), configurability and global view (by having a centralized controller), scalability (the ability to add and dynamically configure network functions as needed), cost efficiency (reduced CAPEX and OPEX), etc.

Both Software Defined Networking (SDN) and Network Function Virtualization (NFV) technologies play a major role in enabling network softwarization and increasing programmability [15], along with other emerging technologies (e.g., cloud computing, edge computing, network slicing) [23].

2.1 Software Defined Networking

The last decade has seen a new wave of networking innovation, largely due to the Software Defined Networking (SDN) paradigm [5]. SDN is currently being deployed in all cloud computing environments and server operations, and is potentially serving as the foundation for the majority of future network services [24]. The key feature of SDN is the separation of the network control plane (responsible for routing) from the data plane (responsible for forwarding) [25]. SDN provides a set of Application Programming Interfaces (APIs) that can implement common network services for business purposes [26].

By decoupling the control plane and data plane, SDN improves the network architecture and eliminates its hierarchy [15]. Network devices (e.g., routers, switches, access points) become forwarding devices. The centralized controller is responsible for managing and controlling all network functions and can dynamically program the network [27] [28]. SDN's centralized architecture provides a faster view of network status, enables more straightforward programmability and updates [5], and has the potential to increase network flexibility and performance [29].

SDN is a key solution to ensure network configurability, global knowledge, and network virtualization [30]. Network virtualization refers to a network that allows service providers to create multiple separate and isolated virtual networks by sharing common physical resources [31].

SDN faces several challenges that need to be addressed for effective implementation. Below, we list the most important ones [5].

- **Controller location:** For instance, SDN introduces an additional communication channel between the data plane and the control plane, which may lack complete transparency, especially in large networks.
- **Scalability:** To avoid bottlenecks, administrators should consider how much control should be delegated to the controller.
- **Security:** This challenges all networks, but SDN-enabled networks are more vulnerable because they are logically centralized.
- **Interoperability:** This is especially relevant in large networks. In addition to device heterogeneity, interoperability between different SDN technologies remains a challenge.
- **Reliability:** This challenge is exacerbated in SDN-enabled networks because the control communication represents a new potential point of failure that should be protected.

In SDN, a single controller is typically responsible for the entire system. To address resulting issues, SDN can distribute the control plane by having multiple controllers [24]. As illustrated in Figure 2.3, SDN approaches can be centralized or distributed [32]. Although the most common case is to have a controller working in a centralized manner (a), it is possible to have a flat distributed controller design (b) or a hierarchical controller design (c).

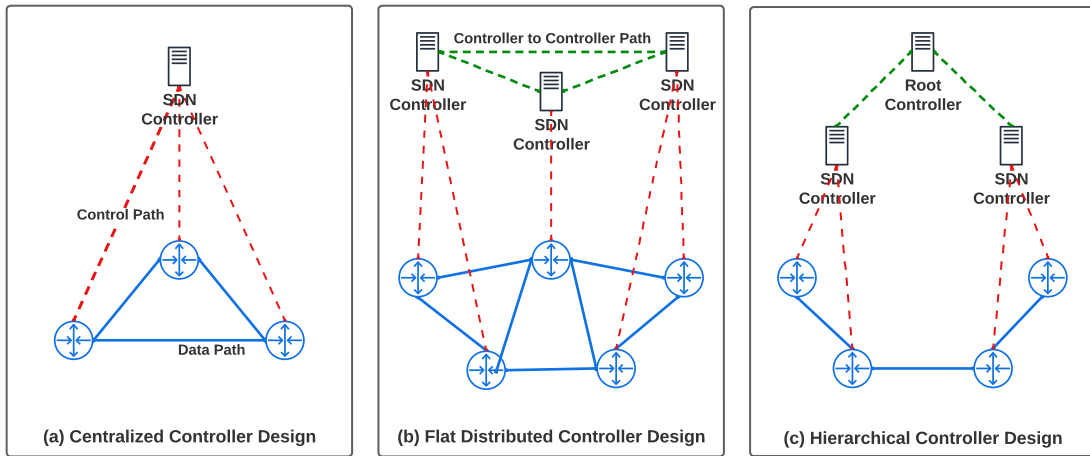


Figure 2.3: SDN Controller Design

The architecture of SDN is depicted in Figure 2.4. It consists of three planes that can communicate through APIs [5] [15] [27].

- **Data plane:** It consists of the set of physical or virtual network devices. It handles incoming frames (i.e., forwarding, modifying, or discarding the frame) according to the policies of the control plane.
- **Control plane:** This is the *brain* of the network, responsible for making decisions and making network management agile. It consists of core network functions that are common to all types of applications (e.g., network topology discovery). Its main component is the logically centralized SDN controller, which provides network abstraction and a global view of the topology and its components.
- **Application plane:** It encompasses business applications designed to meet user requirements and plays a crucial role in defining the network's overall behavior according to the preferences of the network administrator.

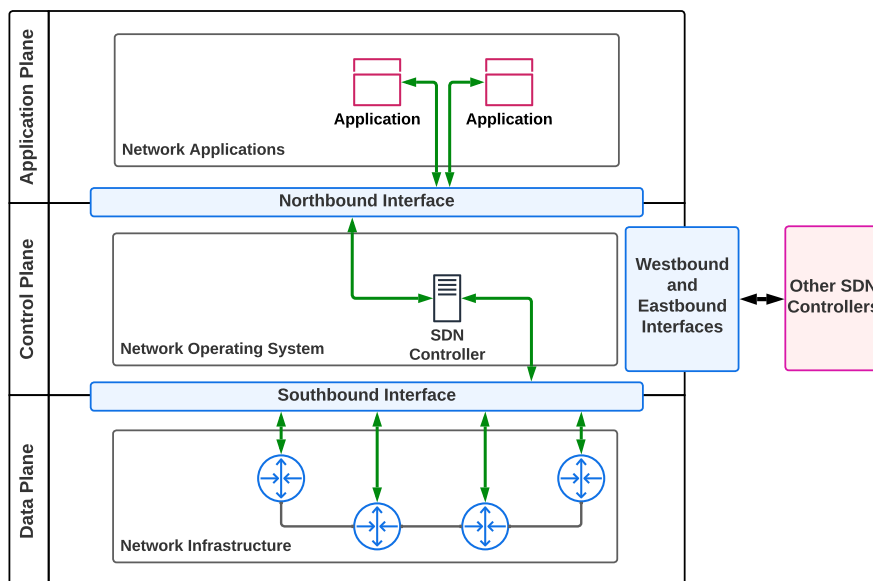


Figure 2.4: SDN Architecture

There are four types of interfaces in SDN that are used to facilitate communication between different layers and multiple components [15]. Southbound interface is used to ensure communication between data plane and control plane. It should be noted that *OpenFlow* is considered as a best practice and example of southbound APIs, as it is used to get state and statistics from forwarding elements and give them control directives [26].

Northbound interface is used as an interface between the control plane and the application plane. It provides the view of the network to the applications. To ensure system redundancy and fault tolerance, it is essential to have multiple SDN controllers that collaborate, make decisions, and communicate with each other via westbound and eastbound interfaces [24].

2.2 Network Function Virtualization

SDN has grown together with the concept of Network Function Virtualization (NFV) as they have joined forces to support emerging network applications [5]. NFV is a key enabling technology for delivering on-demand network services [4].

NFV moves network functions from dedicated hardware to a software virtualized platform. Network devices are hosted in general purpose hardware using virtualization technologies [15]. Thus, network functions become Virtual Network Functions (VNFs) that are implemented on multiple components across multiple virtual machines.

It should be emphasized that NFV has several significant benefits [15]. It enables increased network and resource flexibility, reduced time to introduce new services, improved resource utilization (which implies cost efficiency), and ensuring Quality of Service (QoS) and Quality of Experience (QoE). To achieve these benefits and thus solve most of the current network problems [33], NFV introduces several changes in network service provisioning compared to conventional approaches. In summary, these differences are as follows [28]: decoupling of software from hardware, enabling flexible provisioning of network functions, and facilitating dynamic scaling.

Figure 2.5 introduces the NFV architecture, that is composed of three main layers that are described below [15].

- **Network Function Virtualization Infrastructure (NFVI) layer:** It provides the infrastructure of general-purpose physical devices and virtualization environment. It contains three sublayers: physical infrastructure (i.e., hardware components), virtualization sublayer (i.e., the hypervisor), and virtual infrastructure (i.e., the virtualization environment).
- **NFV Management and Orchestration (MANO) layer:** It is responsible for managing all virtualization processes of NFV. It includes three major categories of man-

agement: virtualized infrastructure manager (manages the resources of NFVIs), VNF manager (manages multiple VNFs), and NFV orchestrator (manages the VNFs lifecycle and orchestrates the resources of NFVIs).

- **Virtual Network Function (VNF) layer:** It is responsible for providing functionalities and services on general purpose.

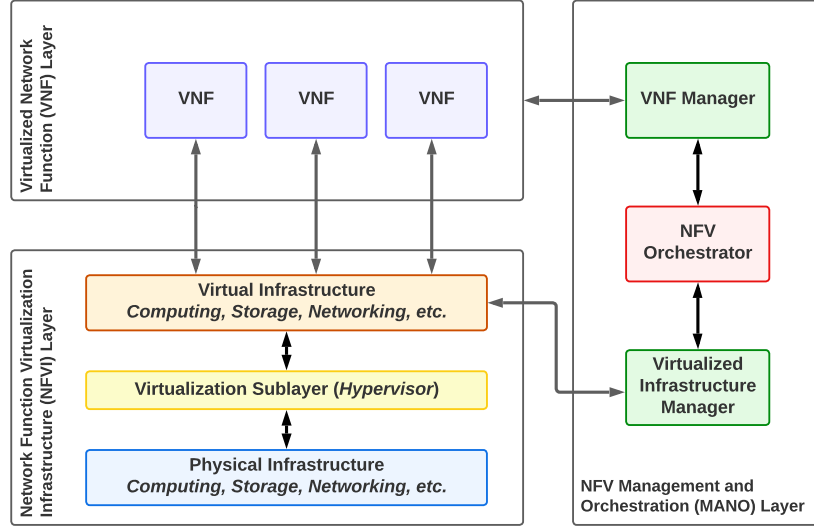


Figure 2.5: NFV Architecture

Admittedly, NFV is still in its nascent phase, as many critical challenges need to be fully addressed [34], and one of the most important challenges is how to optimally allocate virtual resources to network services [1]. Aspects such as management and orchestration, energy efficiency, security, and modeling of resources, functions, and services should also be explored and adequately addressed. It is also important to note that the coexistence of NFV and legacy systems is inevitable before the complete transition to NFV, since it is economically and practically unfeasible to virtualize all functions immediately [22].

3 Machine Learning

Thanks to network softwarization, the application of Artificial Intelligence (AI) and Machine Learning (ML) to networking is easier to implement nowadays [5]. ML is a subset of AI that defines any computational method where the results of previous events or decisions are leveraged to enhance predictions or future decision-making processes [35], allowing systems to improve themselves without being explicitly programmed.

ML approaches typically consist of two main phases, as depicted in Figure 2.6: a training phase and a decision making phase [27]. In the training phase, ML methods are employed to acquire knowledge from a training dataset and construct a system model. In the subsequent decision-making phase, the trained model is utilized to generate estimated outputs for new inputs, facilitating informed decision-making.

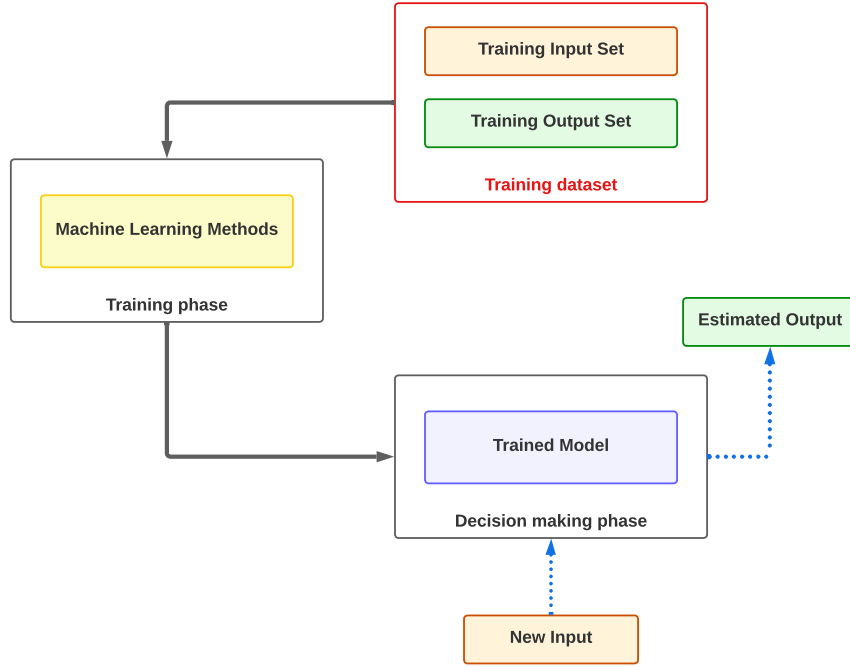


Figure 2.6: Main Phases of an ML Approach

ML is a very large field whose methods have been classified into several categories. We propose a classification approach based on the type of learning involved, thus distinguishing the supervised, unsupervised, and reinforcement learning, and we introduce below many widely used ML algorithms [5] [27]. We also introduce Federated Learning (FL) as an emerging orthogonal learning paradigm that is an alternative to conventional approaches [5].

3.1 Supervised Learning

Supervised Learning (SL) revolves around identifying the unknown function that links the input and output spaces, using labeled input-output pairs for a process known as training. Once the training is complete, the trained model can predict the expected output when provided with new input data.

We present below the most commonly used supervised methods, although certain techniques can also be adapted for unsupervised learning or incorporated into a reinforcement learning framework [5].

- ***k*-Nearest Neighbor (*k*-NN):** Classifies a data sample based on the *k*-nearest neighbors of that unclassified sample.
- **Linear Regression:** Is a straightforward and highly effective ML approach that assumes a linear relationship between the dependent variable and the independent variables.
- **Logistic Regression:** Employs an increasing cost function that is the logistic function rather than a linear one.
- **Markov Decision Process (MDP):** Is a stochastic process operating in discrete time, characterized by the Markov property, which means that the probability of transitioning to a specific state in the next time step depends solely on the current state.
- **Decision Tree:** Employs a learning tree structure where each node represents a feature of the data, branches symbolize conjunctions of features leading to classifications, and leaf nodes correspond to class labels. Unlabeled samples are classified by comparing their feature values with the nodes of the decision tree.
- **Random Forests (RF):** Is a method that aggregates the results of numerous decision trees to estimate a unique value in regression or to determine a class in classification. Each tree contributes a classification result (a vote), and the data sample is classified into the class with the most votes.
- **Artificial Neural Network (ANN):** Consists of interconnected units (i.e., artificial neurons) that use activation functions to perform nonlinear computations and extract knowledge from historical data. Perceptron and Multilayer Perceptron (MLP) were the initial architectures of ANNs.
- **Deep Neural Network (DNN):** Also known as Deep Learning (DL), this category of ANNs encompasses network architectures with a high number of interconnected layers. Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) are two prominent types of DNNs. CNN is a feed-forward network composed of a sequence of cascading convolutional layers. RNN is a stateful network capable of utilizing internal state to handle sequential data.

3.2 Unsupervised Learning

Unsupervised Learning (UL) is designed to identify patterns within unlabeled datasets, eliminating the need for human supervision and pre-labeled input-output pairs. Unsupervised methods autonomously infer relationships between variables based on features such as correlations. These techniques are commonly employed in clustering and data aggregation, grouping sample data into distinct clusters based on their similarities. The most common UL methods are presented below [5] [27].

- ***k*-Means:** Stands as a widely adopted UL algorithm employed to cluster a set of data observations into k clusters. This technique works by minimizing the variance within these clusters. Each observation is allocated to the cluster whose centroid is closest in terms of distance.
- **Gaussian Mixture Models (GMM):** Assumes that observations are generated by a mixture of a finite number of Gaussian variables. This model is probabilistic in nature and extends the k -means approach by incorporating covariance, thereby capturing the uncertainty in cluster assignments.
- **Hierarchical Clustering:** Assembles nearby observations into clusters and establish links. The outcome is a partially ordered dendrogram that gives a hierarchy to the clusters.
- **Self-Organizing Maps (SOM):** Are unsupervised ANNs specifically designed to uncover a low-dimensional discrete representation, known as a map, from the input space. These SOMs are trained using unlabeled data and employ competitive learning principles, wherein only one neuron, termed the Best Matching Unit (BMU), is selected at each step based on the highest similarity between its weight vector and the input.

3.3 Reinforcement Learning

Reinforcement Learning (RL) is an ML paradigm designed to teach an agent (i.e., the learning entity) to make local decisions and take actions to maximize a cumulative long-term reward (via feedback from the environment) [27]. In contrast to SL and UL, RL places a substantial emphasis on the temporal aspect. Consequently, the error metric in RL is distributed over time [5].

As shown in Figure 2.7, the agent monitors the state of the environment and chooses an action at each time step, receives an immediate reward signifying the quality of the action taken, and moves to the next state. The agent's goal is to learn the optimal behavioral policy to maximize the expected long-term reward.

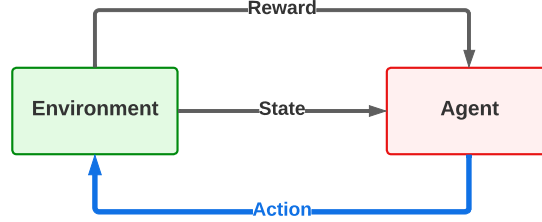


Figure 2.7: RL Typical Scenario

The policy (i.e., the appropriate action for the current state) is a combination of exploration and exploitation. The agent can use exploitation to re-select the action that gave the highest reward in the current state, or can use exploration to select a different action in anticipation of a better reward [36].

In RL, a value function is used to compute the long-term reward of an action given a state. One of the most renowned value functions is the Q-function, utilized by Q-learning to learn a table that contains all state-action pairs and their respective long-term rewards [27], and is responsible for assessing the quality of an action taken in any given state within the system [25]. Q-learning is a model-free RL method to teach an agent how to act according to the state and the observations from the environment. Today, it is the basis for existing RL models.

Deep Reinforcement Learning (DRL) constitutes a subset of RL that leverages the capabilities of DNNs in conjunction with RL models, exploiting the powerful function approximation property of DNNs. Given a state-action pair as input, DRL excels at estimating the long-term reward associated with it. This estimation, in turn, guides the agent in selecting the most favorable action. DRL effectively addresses some of the limitations encountered in traditional RL (e.g., low convergence rate to the optimal action policy, inability to solve problems with high-dimensional state and action spaces).

As RL evolved into DRL, Q-learning evolved into Deep Q-learning [5] by replacing the Markov Decision Support (MDS) framework with DNN, thus solving the problem of multiple states and massive data.

3.4 Federated Learning

As an alternative to conventional ML techniques, Federated Learning (FL) is one of the most attractive techniques that allows a heterogeneous set of devices to train an ML model without sharing the raw data [37] [38] [39]. While improving privacy and communication efficiency, FL also effectively utilizes massive, distributed data and computational resources in constrained networks (e.g., IoT networks) [40]. For example, in FL, only the learning model updates are transmitted between end devices and the FL aggregation server [41].

The FL process generally consists of three steps, as shown in Figure 2.8 [42] [43]: Training task and global model initiation (initially, the central server determines the task requirement and the target application), local model update (each participant trains the model using its local data), and global aggregation (when it receives the local models from the participants, the FL aggregation server performs the aggregation and generates the updated global model).

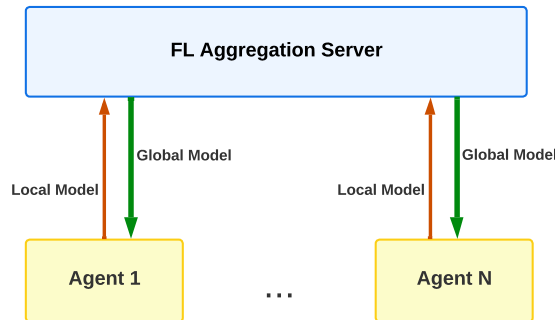


Figure 2.8: FL Typical Scenario

FL algorithms can be classified according to two primary dimensions, namely data partitioning and network structure [39]. In terms of data partitioning, FL can be divided into three categories: Horizontal FL (HFL), where all learning clients collaboratively train a global FL model using local datasets with consistent feature spaces but varying sample spaces, Vertical FL (VFL), where clients share identical sample spaces but differ in feature spaces, and Federated Transfer Learning (FTL) where clients have both distinct sample spaces and feature spaces. Considering the network structure, FL can be divided into two categories: Centralized FL (CFL), characterized by a central server coordinating a group of clients for model training, and Decentralized FL (DFL), where all clients are interconnected in a peer-to-peer (P2P) manner to collectively perform model training.

Note that FL can be combined with RL to form Federated Reinforcement Learning (FRL) [44]. As a new and potential branch of RL, FRL can make learning safer and more efficient while taking advantage of FL. Similarly, DRL and FL can be combined to form Federated Deep Reinforcement Learning (FDRL).

In summary, SL algorithms are generally used to perform classification and regression tasks, while UL and RL algorithms are more suitable for clustering and decision-related tasks, respectively [27]. The choice of the most suitable machine learning method primarily depends on factors such as the dataset nature, size, and the specific problem at hand.

However, the application of ML techniques introduces several challenges that should be considered. For example, training ML models often involves transferring data from its sources, such as IoT devices, to a central system. This approach faces several problems: communication overhead, latency, energy consumption, and privacy concerns [37].

Conclusion

Throughout this chapter, we have examined the background concepts of IoT, network softwarization, and ML techniques. We have outlined the advantages of each technology, thus making the potential of their combination clearer. Next, we will explore the existing work on such combinations.

3

Literature Review of Intelligent Network Softwarization for IoT

Introduction

In IoT networks, the large number of heterogeneous devices poses enormous challenges. This opens up many research areas to build dynamic IoT networks and overcome the limitations. Network softwarization and ML techniques are to be considered to build IoT networks that can overcome the faced problems. In this chapter, after introducing the novelty of our survey, we provide a comprehensive review of the state of the art of the intelligent network softwarization for IoT. We conclude this chapter by discussing the research challenges and future directions.

1 Related surveys

In recent years, IoT, ML techniques, and network softwarization (mainly, SDN and NFV) have received a lot of attention from academia.

In [31], the authors review the state of the art of applying SDN and network virtualization to IoT. They note that prior to their work, there has been no effort to survey the combination of these three technologies. The authors also present a summary of the most prominent architectural frameworks. They identify the following research challenges and future directions: optimizing SDN and *OpenFlow* to adapt to the peculiarities of the IoT paradigm, considering network slicing, considering security challenges, etc. The authors conclude by suggesting to consider the merging of networking and AI.

In [24], the authors review SDN-enabled IoT architectures proposed by different researchers. Their classification of existing architectures leads to the conclusion that there is a need for further research to enhance the integration of IoT with various technologies. This would make IoT solutions more adaptable, scalable, and easier to manage and control. We note that no simulation or evaluation has been done to compare the different architectural approaches. It is recommended to know which integration is better given the constraints of IoT. Authors did not consider NFV technology at this point.

Shamsan and Faridi, authors of the previously cited survey, review in [15] network softwarization for IoT with the adoption of SDN- and NFV-based IoT architecture. Here, SDN centrally orchestrates IoT network flows, while NFV supports the provisioning of IoT network services on demand. The authors discuss their proposed architecture, but do not present its performance evaluation, which prevents comparison with a traditional IoT architecture.

In [6], the authors conduct a review of AI and ML techniques in SDN- and NFV-based networks, categorizing their applications into specific tracks, including network architecture, network planning, network management and operation, and network security. They identify and discuss the challenges and most prominent future directions in the field. The authors point out that the key challenges are: computational complexity and latency, computational and storage resource requirements, access to resources and datasets (there is a lack of openly available standard network datasets), and storage of valuable data (there is no trend to store communication data for future use). Conversely, we note that the authors did not execute the studied methods.

In [45], the authors review and examine the opportunities and challenges of adaptive and data-driven softwarized networks, and introduce a conceptual framework for adaptation (i.e., flexible response to new requirements and evolving contexts due to the diversity of applications) in softwarized networks. The data-driven decision modules (i.e., ML modules) can learn and make informed decisions in response to environmental changes, thereby facilitating meaningful decision-making for adaptation in softwarized networks. We note that the authors did not address the IoT scenario.

In [27], the authors provide a comprehensive review of the literature on ML applied to SDN from the perspective of traffic classification, routing optimization (where the use of RL outperforms the conventional heuristic algorithms), Quality of Service (QoS) and Quality of Experience (QoE) prediction, resource management, and security. The authors also discuss the challenges, mainly mentioning the need for high-quality standardized datasets. In summary, the research on the application of ML algorithms is quite broad and many challenges lie ahead. Even if the authors compare the performance of ML-based solutions in SDN in terms of accuracy, they could not evaluate them directly, as different learning algorithms use different datasets for training and simulation is not performed under the same conditions.

While reference [27] provides a general survey, the authors in [5] provide a more specific survey, focusing on the application of ML for optimizing routing in SDN-enabled environments. They highlight the ongoing need for more extensive comparisons and collaborations among different approaches, as well as the necessity for meaningful evaluations based on openly available datasets and network topologies. Additionally, the authors acknowledge the challenge of acquiring large datasets and suggest addressing it by initially training models with synthetic data followed by fine-tuning using smaller real datasets. The authors conclude that DRL is particularly relevant in the last two years, as most of the published works fall into this type of ML technique. However, there is a lack of efforts to create synergies or to compare different ML approaches for SDN routing. Many evaluations only compare their methods with traditional routing protocols and not with competing proposals, likely due to limited public availability of implementations. This lack of reproducibility and comparability hinders the meaningfulness and conclusiveness of these evaluations.

In [46], one of the rare surveys that focuses on the combination of ML, SDN, NFV, and IoT (although it does not cover all ML approaches, contrary to ours), the authors present

an overview of using DRL for managing SDN and NFV in Edge-IoT scenarios. It is worth highlighting that Edge Computing (EC) architecture plays a pivotal role in delivering quicker service response times and cost reduction in processing IoT data compared to traditional cloud-based approaches. As observed by the authors, the surveyed works often lack a primary focus on RL or DRL solutions, or provide only superficial coverage.

In [47], the authors discuss the traditional IoT networks and the need for SDN and NFV to address IoT challenges. The authors acknowledge that a lot of research has been done in the areas of IoT, SDN, and NFV, but they note that there is a missing link in terms of the evolution of IoT architectures, from primitive frameworks to sophisticated SDN- and NFV-enabled platforms. Authors attempt to fill this gap. They superficially discuss the application of AI and ML techniques in IoT (i.e., cognitive IoT networks) and in IoT network softwarization. The authors explore the architectural evolution of IoT. They also survey the state of the art of the available IoT simulators.

Table 3.1: A Brief Comparison of Our Survey with Existing Ones

Year	Paper Type	Ref.	Network		ML				IoT
			Softwarization		Technique(s)				
			SDN	NFV	SL	UL	RL	FL	
2016	Journal	[31]	✓	✗	✗	✗	✗	✗	✓
2018	Conference	[24]	✓	✗	✗	✗	✗	✗	✓
2019	Journal	[15]	✓	✓	✗	✗	✗	✗	✓
2019	Conference	[6]	✓	✓	✓	✓	✓	✗	✗
2019	Journal	[45]	✓	✓	✗	✗	✗	✗	✗
2019	Journal	[27]	✓	✗	✓	✓	✓	✗	✗
2021	Journal	[5]	✓	✗	✓	✓	✓	✗	✗
2021	Workshop	[46]	✓	✓	✗	✗	✓	✗	✓
2022	Journal	[47]	✓	✓	✗	✗	✗	✗	✓
		Ours	✓	✓	✓	✓	✓	✓	✓

As summarized in Table 3.1, we find that although many research efforts have been made to review the three promising technologies (IoT, ML, and network softwarization) and their combination, only a few of them have considered combining all these three tech-

nologies, but within a narrow vision (e.g., considering only one ML technique). Moreover, to the best of our knowledge, no existing work has really focused on ML-enabled network softwarization for IoT networks as a promising solution to address multiple challenges. It is also to note that there is no work that consider FL technique as an enabler for efficient network softwarization, especially for IoT networks. To fill this gap, we provide a comprehensive overview of the intelligent network softwarization for IoT networks.

2 Literature review

In this section, we review the state of the art of intelligent network softwarization for IoT. First, we introduce the existing applications of network softwarization for IoT. Then, we highlight the existing work on ML-enabled network softwarization. Finally, we review the state of the art of intelligent network softwarization for IoT.

2.1 Network softwarization for IoT

In IoT, network scalability and flexibility are critical. SDN and NFV are suitable network softwarization technologies to enable these functionalities. This topic has received a lot of attention from the research community. In the following, we present some relevant papers on this topic.

In [30], the authors tackle the challenge of integrating SDN architecture, which can introduce overhead that potentially affects overall network performance, into IoT networks, specifically focusing on IEEE 802.15.4 low-power wireless networks. They introduce a lightweight SDN framework, μ SDN, and evaluate it in terms of latency, energy, and packet delivery. The simulation results demonstrate that their proposed solution maintains scalability in comparison to a traditional IEEE 802.15.4 network employing the Routing Protocol for Low-Power and Lossy Networks (RPL) routing, while allowing the network to benefit from SDN. It significantly reduces delay and jitter in an interference scenario where traditional approaches face difficulties. Note that the authors do not apply NFV in conjunction with SDN, even though they note that SDN enables NFV.

Considering the issue of energy consumption in IoT networks, the authors in [26] propose a novel SDN-based routing technique for IoT that allows different devices and resources to connect to each other with minimum energy consumption. Necessary mea-

measurements have been performed and the power consumed in each sensor has been measured. The authors simulate the proposed work with *NS2* network simulator and present a performance analysis (simulation parameters are mentioned). The metrics used are throughput and Packet Delivery Ratio (PDR). However, even though the authors assume that their solution improves the network performance, no comparison with other solutions (e.g., the commonly used RPL routing protocol) is made.

In [48], the authors propose an energy-aware SDN- and NFV-enabled architecture for IoT. The authors argue that data aggregation (which can be defined as an NFV instance and dynamically deployed to the required IoT nodes) can further reduce network traffic and improve the overall resource utilization. The solution makes appropriate changes and optimizations to various network functions to cope with IoT networks. After detailing the proposal, the authors proceed to a simulation using *Cooja*. Extensive evaluation confirms that the proposal outperforms its counterparts in terms of energy usage and PDR. It should be noted that the proposed solution is based on the μ SDN as proposed by Baddeley et al. in [30].

In [12], the authors explore the potential of merging blockchain and SDN to address certain challenges within IoT networks. They propose a secure and energy-efficient blockchain-enabled architecture of SDN controllers for IoT networks. The architecture involves a cluster structure and introduces a novel routing protocol. Through experimentation, the authors demonstrate that their proposed routing protocol outperforms existing counterparts in performance and energy efficiency. We note that the authors omit the well-known RPL protocol when comparing the proposal with its counterparts.

In [49], the authors introduce the concept of virtual IoT honeynets as a strategy for countering cyberattacks in softwarized IoT networks. Enhanced with SDN and NFV support, honeynets operate as virtualized services, replicating real IoT network configurations to divert potential attackers away from actual targets. The authors proceed to implement and assess their solution, showcasing its effectiveness in mitigating cyberattacks. The deployment of these honeynets relies on user-defined security policies. As future research, we suggest to consider ML techniques and especially RL to make the system automatically define and improve its security policies.

In [50], the authors introduce a security orchestration framework that is both semantic-aware and policy-driven. This framework is designed for orchestrating security measures autonomously in softwarized IoT networks, with a specific focus on ensuring conflict-free

security orchestration while optimizing the allocation of security VNFs. The proposal has been validated in a real testbed, demonstrating its feasibility and performance in detecting conflicts. As future work, we propose to consider incorporating intelligence into the solution to enable self-driven security orchestration.

The authors in [51] design an SDN- and NFV-enabled IoT network architecture that considers resources and energy constraints as the main priority. They extensively evaluate the performance of their solution using *Cooja*, and the solution outperforms its counterparts in terms of PDR, network lifetime, and overhead reduction. This paper extends the previously cited reference [48] by presenting a novel optimization model. We are pleased to note that the authors provide the prototype implementation code for future reproduction and extension, a practice that we strongly encourage to stimulate research efforts in this area. As future work, they suggest considering a real testbed implementation for a better evaluation of the proposal. What we suggest as a future direction is to consider ML techniques to propose an intelligent solution of the optimization problem.

In [3], the authors propose a novel mechanism for load balancing routing and virtualization in IoT networks using SDN. The authors recall that load balancing has four main goals: resource utilization, QoS, resilience, and scalability. Their approach leverages the *OpenFlow* protocol to directly monitor link load information and network operational status, allowing for load balancing routing decisions to be tailored to individual flows across various IoT applications. After introducing the proposed algorithm, the authors evaluate its efficiency through simulation using *NS2*. It has been compared with existing algorithms, and the simulation results support the findings. The simulation was performed for different use cases with different requirements. Note that this solution assumes a static network topology without mobile nodes, while real-world IoT networks often feature dynamic elements, including mobile nodes.

In [4], the authors propose a novel architecture to overcome the limitations posed by resource constraints in IoT through softwarization. The authors note that while existing works on softwarization within the IoT realm have tended to be specialized, focusing on solving specific problems, their proposed architecture takes a broader approach. It is designed to be a versatile solution aligning with the standard architectures of SDN, NFV, and IoT. The proposal is built and simulated using the *Mininet* network emulator. The experimental outcomes show that there is a significant improvement in the performance of the proposed architecture over a conventional IoT system.

In [52], the authors investigate various SDN- and NFV-based architectures tailored for edge-cloud-oriented IoT environments. Their primary objective is to assess the feasibility of deploying, administering, and disseminating network services to end-users within these contexts. They highlight that, so far, challenges linked to communication between edge and cloud components and the provision of network services in IoT networks have not been adequately addressed. However, no simulation of the architectures was conducted, making it difficult to consider the effectiveness of the proposal.

In conclusion, nowadays a great effort is being made to fully softwarize the IoT networks through the SDN and NFV technologies, which is confirmed by the reviewed papers, thus addressing the main challenges of the IoT. However, we point out that these references do not include intelligence in the solution. Intelligence, via ML techniques, is certainly a key feature to ensure an efficient network softwarization for IoT.

2.2 ML-enabled network softwarization

Today, ML techniques are a driving force of several domains, and it is certainly a booster for network softwarization. They have the potential to effectively address many network-related challenges, including design, implementation, performance optimization, and verification. Therefore, it is quite normal that academia has focused on the topic of enabling ML for SDN and NFV in recent years, as presented below.

In [36], the authors propose an RL-based algorithm for selecting an appropriate path for a Service Function Chain (SFC - i.e., a sequential chain of VNFs) in an SDN- and NFV-enabled network. The proposed solution selects an appropriate path depending on the network conditions, thus ensuring an efficient service chaining environment. The authors chose RL techniques because they believe that it is the most appropriate for decision making problems. The reward depends on computational and bandwidth usage. The authors implement a simulator with *Java* and evaluate the performance of their proposal. It outperforms its counterparts. However, we note that the authors did not consider constrained networks and thus did not consider an energy-related metric.

In [53], the authors present a DRL-based framework tailored for online end-user service provisioning in an NFV-enabled network. They pose an optimization problem with the goal of minimizing the cost of network resource utilization, and introduce a Deep Q network as a solution to this optimization challenge. In particular, the authors provide

an assessment of the computational complexity associated with their proposed approach. Furthermore, to measure the performance of their framework, the authors benchmark it against various baseline methods. The results, derived from various parameter settings, show the effectiveness of the framework.

In summary, many research efforts have been made to merge ML with network softwarization and mainly SDN and NFV, but we note that authors usually consider SDN as the main key enabler for network softwarization and omit NFV, although they are mutually enabling. We also note that even if some works consider different network types, they omit IoT networks, which are continuously growing and require efficient intelligent network softwarization techniques.

2.3 Intelligent network softwarization for IoT

ML plays an essential role in creating smarter IoT networks by enabling them to learn from experience, making them more resilient to vulnerabilities and failures while improving overall performance. As the architecture of network softwarization enables easy integration of ML, we strongly believe that we will soon see a fusion of ML and network softwarization techniques with IoT. In the following, we present the most recent and relevant work, although there is a small number of publications on the topic.

In [25], the authors present an energy-efficient clustering and routing algorithm tailored for monitoring large-scale IoT networks based on SDN technology. They present a novel modeling approach structured around a dual-level control mechanism to address the challenge of the lack of an efficient routing protocol capable of handling a significant number of devices while ensuring low-power data forwarding. The authors aptly recognize that current algorithms are ill-suited for efficient routing in large-scale networks. These algorithms do not effectively scale to large device populations, nor are they energy efficient, especially in scenarios involving mobile devices. To address this problem, the authors employ a two-tiered approach: at the first tier, they implement a multi-hop clustering routing mechanism in conjunction with RPL routing, while at the second tier, they employ an SDN controller in combination with a Q-routing algorithm based on Q-learning. This two-tier setup enables intelligent network management on a global scale. To evaluate the effectiveness of their model, the authors perform simulations using the *Cooja* platform. The results clearly show that their proposed model delivers significantly better results compared to current state-of-the-art solutions.

In [7], the authors propose a DRL approach for energy-efficient task assignment and scheduling in an SDN-based fog IoT network. Task scheduling is a notoriously challenging problem that often involves complex online decision making. The primary goal of this approach is to minimize network latency while ensuring optimal energy efficiency. By integrating DRL, the method leverages intelligent agents capable of learning and making better decisions through direct interaction with the environment. The authors use a testbed setup to evaluate and analyze the performance of their proposed approach. The evaluation results show its superiority over both deterministic and random task scheduling strategies. However, it is worth noting that certain data-related challenges, such as privacy concerns, need to be addressed, and FL emerges as a promising enabling technology to effectively address these issues.

In [35], the authors highlight the significant problem posed by the uncontrolled proliferation of insecure IoT-based devices. They outline the vulnerabilities associated with these devices and propose a solution using an NFV infrastructure in conjunction with emerging technologies to enable intelligent management and enhanced protection against botnet attacks. Intrusion attacks, particularly in the form of botnets, are a common threat in IoT networks. Botnets consist of a large network of compromised devices acting as bots and are often associated with Distributed Denial of Service (DDoS) attacks, Man in the Middle (MitM) attacks, malware distribution, and other security risks. IoT botnets are of particular concern due to their rapid propagation, which allows them to have a greater impact than traditional botnets. The authors present a potential NFV architecture that incorporates several emerging technologies into the solution. These technologies include a virtual honeynet that acts as a decoy to trap attackers, cloud computing to reduce latency in providing protection against vulnerabilities, and ML to analyze cyber threats. While the authors provide a detailed description of their proposed solution, it is important to note that it has not been implemented or evaluated for performance, particularly in real-world scenarios involving known and zero-day attacks.

In [2], the authors present a novel approach involving a modified DRL agent designed for dynamic resource placement within IoT network slicing. This specialized agent engages with controllers and orchestrators to effectively manage the installation of flow rules and the allocation of physical resources in NFVI. The approach introduces a unique formulation that incorporates completion time and criticality criteria. To evaluate the proposed method, the authors conduct simulations using the *Mininet* emulator. They assess performance using key metrics such as delay and PDR. The comparative analysis

involves three agents: the proposed modified DRL agent, an unmodified DRL agent, and an experience-based allocation agent. The results of the evaluation demonstrate that the proposed solution surpasses its counterparts in terms of performance.

While most existing works assume that services are represented as SFCs, which are chains (for reference, we previously reviewed the matter in [36]), the authors in [1] consider that network services in IoT networks exhibit greater complexity and diversity. Consequently, they propose a more suitable representation known as VNF Forwarding Graphs (VNF-FGs), which are essentially Directed Acyclic Graphs (DAGs). They point out that prior research has failed to fully leverage this specific graph structure, rendering them suboptimal or unsuitable for IoT networks. In light of this, the authors delve into the VNF-FG placement problem within dynamic IoT networks. To effectively exploit the graph structures of services and address the challenges of dynamic IoT networks, they employ a combination of Graph Neural Network (GNN) and DRL. Their resulting algorithm, named *Kolin*, is designed for efficiency. Extensive simulation results demonstrate that *Kolin* outperforms existing solutions.

In [54], the authors study an ML routing protocol in mobile SDN-based IoT. The authors note that RPL, one of the prominent routing protocols in IoT, assumes that all nodes are static. Therefore, this routing protocol does not provide a mechanism to support mobile nodes. To overcome this problem, they propose to combine ML and SDN. Unfortunately, the solution is not detailed, and no simulation is performed to evaluate the performance based on relevant metrics.

To address edge FL challenges in large-scale heterogeneous IoT networks, the authors in [55] introduce a model that integrates SDN and NFV. This integration facilitates the deployment of NFV-enabled edge FL aggregation servers, enhancing automation and control. The proposed solution is powered by ML, specifically utilizing Multi-Agent Deep Q-Networks (MADQNs) to enable self-learning through softwarization. The authors conducted simulations to evaluate the performance of their solution, and the results demonstrate that it outperforms reference methods in terms of QoS metrics.

In [29], the authors introduce a smart and energy-efficient traffic flow control strategy for SDN-based IoT networks. Their proposal aims to identify the optimal set of border nodes for each cluster within the network, which helps reduce the overall number of potential paths between clusters. The selection of these optimal border nodes is based on criteria involving maximum energy and minimum distances. To identify this set

of optimal border nodes, the authors employ the Lion Swarm Optimization Algorithm (LSOA), a nature-inspired optimization technique. Although LSOA is not a machine learning method, it is a recent and noteworthy approach in this field. The simulation results conducted using *MATLAB* indicate that border nodes selected using LSOA outperform those selected using other state-of-the-art metaheuristic algorithms. This leads to enhanced network efficiency through energy conservation. As a future research direction, ML algorithms could be considered.

In [56], the authors introduced an ML approach to place security VNFs (e.g., firewall) to mitigate DDoS attacks on industrial IoT. The authors note that few works have used NFV to detect and mitigate threats on industrial IoT networks, and even fewer have considered network performance indicators when placing VNFs. The proposed approach is novel as it considers NFV performance indicators as decision variables (e.g., deployment time, computation resource utilization, and memory consumption). Experiments show that ML is an effective alternative, as it shows 99.40% accuracy with respect to ideal placement. The authors have considered and compared several ML techniques (e.g., k -NN, RF). It is good to note that the authors made all the produced code and data publicly available, which allows the community to reproduce the experiments. We note that the authors considered only one specific attack, but their approach could be applied to mitigate other types of attacks, such as MitM attacks.

Finally, we note that even though some efforts have been made recently to merge ML, network softwarization (mainly SDN and NFV technologies), and IoT, as described above (the number of research papers mainly increased from 2022), more efforts are still needed to achieve effective solutions.

3 Discussion and future directions

It is evident that academia is fully aware of the potential of SDN, NFV, ML, and IoT as key enabling technologies. Each technology has received a lot of attention on its own, and as a result, recent research papers have proposed to merge them. Since network softwarization (which includes SDN and NFV technologies) is essential to ensure efficient, agile IoT networks, and since ML algorithms are more appropriate for softwarized networks than traditional ones, we are moving toward effective intelligent network softwarization for IoT very soon.

Table 3.2: Advances on the Combination of Network Softwarization, ML, and IoT

Ref.	Network		ML Tech- nique(s)	IoT	Solution Evalua- tion	Objective(s)
	Soft-warization					
	SDN	NFV				
[30]	✓	✗	✗	✓	✓	Evolving SDN to be adapted to constrained IoT networks.
[26]	✓	✗	✗	✓	✓	Improving routing protocols for IoT networks.
[48]	✓	✓	✗	✓	✓	Providing an energy aware soft-warized architecture for IoT.
[12]	✓	✗	✗	✓	✓	Proposing an energy-aware SDN architecture for IoT with blockchain-based security.
[49]	✓	✓	✗	✓	✓	Proposing virtual honeynets to counter cyberattacks in soft-warized IoT networks.
[50]	✓	✓	✗	✓	✓	Proposing a semantic-aware security orchestration framework in softwarized IoT systems.
[51]	✓	✓	✗	✓	✓	Proposing a resource-aware IoT network architecture based on SDN and NFV.
[3]	✓	✗	✗	✓	✓	Proposing a mechanism for load balancing routing and virtualization for SDN-enabled IoT.
[4]	✓	✓	✗	✓	✓	Proposing a novel IoT architecture based on network soft-warization.

Continued on next page

Table 3.2: Advances on the Combination of Network Softwarization, ML, and IoT
(Continued)

Ref.	Network		ML Tech- nique(s)	IoT	Solution Evalua- tion	Objective(s)
	Soft- warization					
	SDN	NFV				
[52]	✓	✓	✗	✓	✗	Presenting SDN- and NFV-based architectures for edge-cloud oriented architectures.
[36]	✓	✓	RL	✗	✓	Studying RL-based SFC path selection in softwarized networks.
[53]	✗	✓	DRL	✗	✓	Studying a DRL based framework for online service provisioning in NFV-enabled networks.
[25]	✓	✗	Q-learning	✓	✓	Providing an energy efficient clustering and routing algorithm for SDN-based IoT networks.
[7]	✓	✗	DRL	✓	✓	Presenting a SDN-based dynamic task scheduling and resource management DRL approach for IoT.
[35]	✗	✓	✓	✓	✗	Proposing an NFV-based strategy for enhanced protection against bot attacks in AI-driven IoT.
[2]	✗	✓	DRL	✓	✓	Proposing a DRL agent for dynamic resource placement in IoT network slicing.

Continued on next page

Table 3.2: Advances on the Combination of Network Softwarization, ML, and IoT
(Continued)

Ref.	Network		ML Tech- nique(s)	IoT	Solution Evalua- tion	Objective(s)
	Soft- warization					
	SDN	NFV				
[1]	✓	✓	GNN, DRL	✓	✓	Proposing a GNN-assisted DRL method for VNF-FG placing in softwarized IoT networks.
[54]	✓	✗	✓	✓	✗	Studying ML-based routing protocol in mobile SDN-enabled IoT networks.
[55]	✓	✓	✓	✓	✓	Proposing a Multi-Agent Deep Q-Networks for improved edge FL communications in softwarized IoT.
[29]	✓	✗	LSOA (not a ML tech- nique)	✓	✓	Proposing an intelligent energy efficient method for managing traffic in SDN-powered IoT networks.
[56]	✗	✓	✓	✓	✓	Proposing an ML method to position security VNFs to combat DDoS attacks in industrial IoT.

However, after examining the most relevant and recent research works on the topic, which we summarize in Table 3.2, we conclude that the works that exploit the full potential of the cited technologies in conjunction are still sparse. We observe that many works that consider network softwarization only consider SDN and omit NFV, even if the combination of both ensures the efficiency of the softwarization. There is an important progress in intelligent softwarization, but it is not always applicable in IoT networks.

It is also important to point out that few works propose a consistent implementation and evaluation, due to lack of access to the necessary resources (e.g., network datasets,

source codes). We concur with all authors who strive to have high quality normalized network datasets to be able to implement solutions and perform coherent performance evaluations. Having access to the code of the proposals and the simulation parameters is essential to be able to make comparisons and thus improve the proposals. Otherwise, research in this emerging area will be limited, but also scattered, as authors will not be able to improve and build on existing work. Having reproducible research is the key to real implementation in practice, to industry-based practical solutions.

We aim to have more publications on the integration of IoT with intelligent network softwarization. Obviously, the latter should be adapted to the peculiarities of IoT networks. Therefore, authors should consider the IoT constraints (e.g., energy limitations) when setting the evaluation metrics. It is important to make consistent comparisons, as we found that authors usually compare their proposals to conventional solutions, not to the smart counterparts.

It is undeniable that performing simulations to evaluate the proposal is very important for the development of this promising field, so the choice of the simulation platform is very important for the veracity of the results. In the early stages, the implementation can be done using a simulator, an emulator, or a testbed [57]. A simulator is the right choice in the first phase of the design because it provides a higher level of abstraction. Emulators map real-world devices to simulated ones, so the results are more reliable than those of simulators. Testbeds are much more accurate than software-based tools because they are a hardware-based tool for implementing and evaluating a solution.

As future research directions, we suggest focusing on new emerging technologies such as Edge Computing (EC). This technology offers the benefits of faster service response times and lower data processing costs for IoT applications by shifting computationally intensive tasks from less capable IoT devices to more powerful edge servers. It is also interesting to consider a distributed multi-controller softwarization architecture, where multiple controllers coexist in a clustered multi-tier scheme, thus ensuring scalability and reliability while avoiding Single Point of Failure (SPoF). Having such a distributed architecture implies the possibility to propose distributed learning approaches (i.e., FL).

Finally, to ensure real implementable solutions, security should be given the necessary attention, as it is a cross-cutting aspect in the intelligent network softwarization of IoT. It is a vast domain and a lot of research could be done towards a self-secured IoT.

Conclusion

Considering the current emerging technologies, intelligent network softwarization is certainly the enabler to address the constraints and challenges of IoT. After analyzing the recent advances in the area, we have concluded that the works exploiting the full potential of the combination of the three promising technologies (i.e., IoT, network softwarization, and ML) are still sparse. We have also demonstrated the novelty of our review work compared to related surveys in this area, and we have made some publications based on the findings of this literature review. Next, as we have identified the most prominent future directions, we will consider these technologies to propose an intelligent distributed network softwarization architecture for IoT.

4

Intelligent Distributed Network Softwarization for IoT

Introduction

As IoT use cases keep evolving, networks face tremendous challenges that need to be addressed by incorporating the most prominent emerging technologies. In this chapter, we present our contribution to fill the gap by proposing an intelligent distributed network softwarization architecture for IoT, leveraging ML techniques and network softwarization. After introducing our proposed architecture and the main technologies, we propose and thoroughly describe a Federated Deep Reinforcement Learning (FDRL)-based routing solution in softwarized IoT environments. This solution, based on our proposed general architecture, will take advantage of intelligence and softwarization to meet the requirements of IoT networks in terms of QoS and energy efficiency, and can be further enhanced using NFV.

1 Proposed architecture

Here, we present a novel enabling architecture to achieve intelligent distributed network softwarization for IoT networks. The proposal was build based on the state-of-the-art architectures of IoT (cf. Figure 2.2), SDN (cf. Figure 2.4), and NFV (cf. Figure 2.5), as presented earlier. We first present an overview of the solution, followed by a presentation of the technologies.

Based on the conducted state-of-the-art review, we propose the following architecture. Considering the conventional IoT architecture, which mainly consists of sensors and controllers, and to mitigate the IoT network issues, we add SDN and NFV components, as these are the main network softwarization techniques, to obtain an IoT network softwarization architecture.

The number of IoT devices is growing exponentially, and therefore, a single SDN controller driving an entire IoT network will become an issue in terms of reliability. Therefore, unlike other proposals in the field, we opt for a distributed hierarchical SDN controller design (cf. Figure 2.3), where multiple controllers coexist and are coordinated by a root controller. Such a distributed design allows the application of distributed learning approaches, namely FL (cf. Figure 2.8). It will allow the system not only to take advantage of ML for autonomous efficient decision making, but also to build a global knowledge without having to share raw data between the controllers, as each controller acts as a learning node, and where the root controller acts as an FL aggregator.

Figure 4.1 shows the proposed architecture, which consists of five layers combining the IoT, SDN, NFV, and FL components. The presence of FL components will allow the incorporation of various ML techniques to add intelligence to the network, such as DRL techniques, which have shown excellent results in online decision problems (like the majority of problems in networking). In the following, we present the proposed architecture layers:

- **Application Layer:** Contains the applications used to process and manage the data to be useful to the end user, as it is the front end of the whole system. Through the service applications, the user can request services that are provided by the IoT applications, SDN applications, and VNFs (as they make the requests understandable to the underlying layers).

- **Global Knowledge and Management Layer:** Centralizes network management and control through the head root SDN controller. This controller also acts as the FL aggregator, providing global intelligence by generating and broadcasting the global model to the multiple SDN controllers, and ensuring coordination with the IoT controller and NFV hypervisor.
- **Control Layer:** Contains the controllers of each domain (IoT, SDN, and NFV). Each monitors the underlying devices. SDN controllers contribute to global intelligence by acting as FL learning nodes.
- **NFV Management and Orchestration (MANO) Layer:** Contains the management components of the network functions virtualization system that manages all of the virtualization processes of NFV.
- **Physical Layer:** Contains the physical resources of the system, including IoT sensing devices, network forwarding devices, and the infrastructure of the NFV environment.

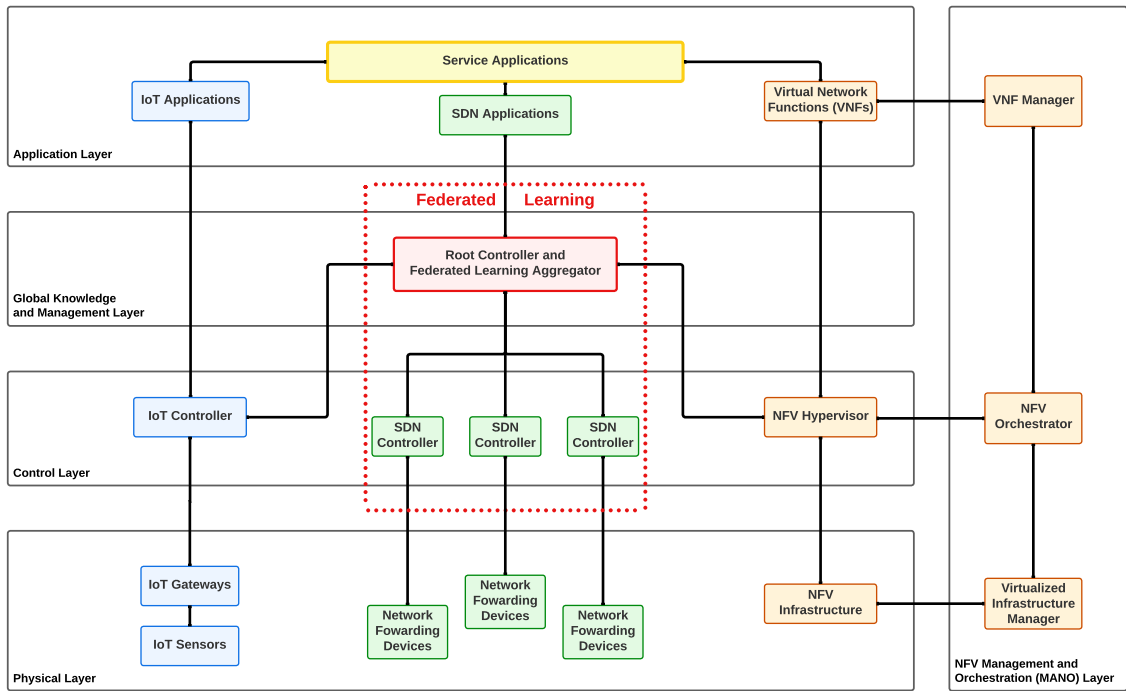


Figure 4.1: Intelligent Distributed Network Softwarization Architecture for IoT

The proposed architecture is based on technologies that have proven their effectiveness in addressing a variety of challenges. By being general, it is intended to be the basis for future specific work. For example, it can be used to address security concerns by using NFV to dynamically place security functions, and intelligent SDN to make networks self-secure.

2 Routing optimization in softwarized IoT

As IoT networks grow, they become increasingly dynamic and difficult to model and control. Traditional routing mechanisms, such as Shortest Path Routing (SPR), are unable to meet the requirements in terms of QoS, but also in terms of energy efficiency. In this section, we present our proposal for routing optimization in softwarized IoT, based on the general architecture presented earlier. This solution takes full advantage of distributed intelligence by being based on FDRL, which combines the forces of DRL and FL.

2.1 Solution overview

Consider an IoT network where any node can communicate with any other node at any time. Each flow has specific requirements on a set of QoS metrics (e.g., latency, throughput, PDR). Since we are evolving in a constrained environment, each hop implies an energy consumption (due to data transmission and processing at intermediate nodes). Therefore, from an energy efficiency point of view, it is better to reduce the number of hops. Our goal is to dynamically decide which route to take for each communication flow. The flow should be successfully forwarded from the source to the destination while satisfying the quality and energy requirements.

Since we are in a distributed softwarization architecture, each SDN controller is responsible for a part of the network. DRL is considered as the most appropriate technique for online decision problems, of which routing is one [58] [59]. Our architecture allows us to evolve towards FDRL, which allows controllers to have stronger routing policies as they are trained on larger datasets without having to exchange raw data, which would have increased communication overhead and opened up potential security failures. In table 4.1, we propose a comparison of SPR, DRL routing, and FDRL routing.

Much work has been done and proven the effectiveness of DRL routing, and FDRL routing will surely replace its counterparts once all challenges have been addressed, as it is the most complicated approach (due to the need to handle both FL and DRL complexities). This work is the baseline to have a fully functional routing solution in the near future, especially if research forces are combined.

Table 4.1: Comparison of SPR, DRL, and FDRL Routing Approaches

	SPR	DRL routing	FDRL routing
Implementation complexity	Low (e.g., Dijkstra)	High (DNN)	Extremely high
Need for training	No	Yes	Yes
Real-time adaptability	Static	Highly adaptive	Highly adaptive
Multi-objective	No	Yes	Yes
Scalability	Low (good for small networks)	Good (scales well with network size)	Excellent (scales across multiple domains)
Privacy	Not considered	Not considered	Designed for data privacy

In Figure 4.2, we present an overview of the routing solution. It is consistent with the general network softwarization architecture proposed earlier. We recall that routing is one of the multiple SDN applications. It periodically gathers network statistics to ensure that the selected path for a source-destination pair request remains consistent with the current network state.

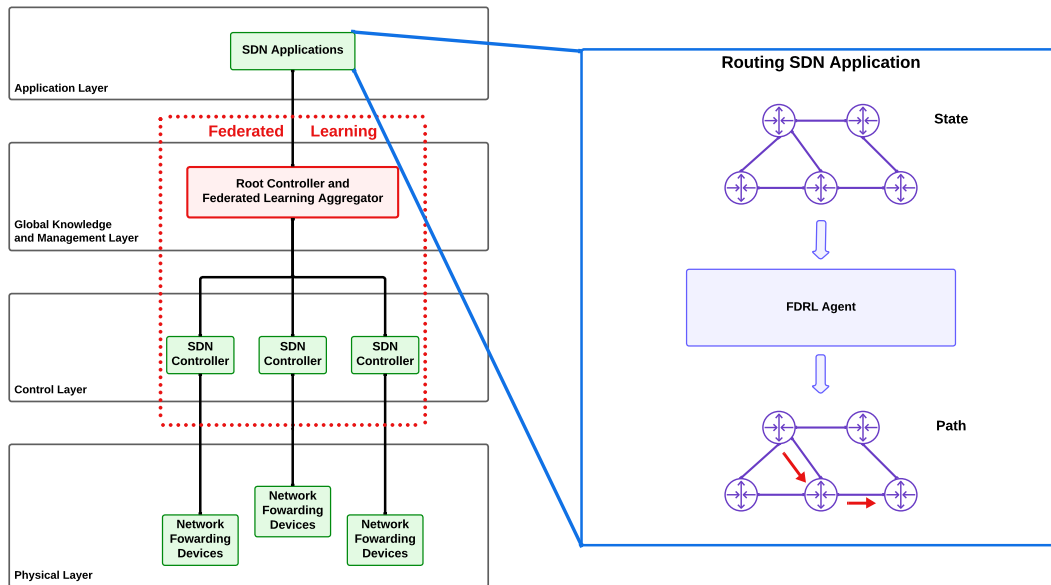


Figure 4.2: FDRL Routing for Softwarized IoT

2.2 Problem formulation

Formally, we represent the network as a directed graph $G(V, E)$, where V denotes the set of switches and routers, and E denotes the set of links between them, such that $E = \{(i, j) \in V \times V, i \neq j\}$. For each link $(i, j) \in E$, the delay (i.e., latency), loss, and bandwidth metrics are represented by $\{d_{ij}, l_{ij}, b_{ij}\} \in \mathbb{R}_+$, respectively.

Given a source node α with a flow f that has as destination node β , our goal is to find a path that minimizes delay and loss while maximizing throughput, thus ensuring good QoS. Since we also need to improve the energy efficiency, we should reduce the number of hops and thus the energy consumption for the flow transmission. We refer to the number of hops of a given flow f from α to β as $h_{\alpha, \beta, f}$. This metric can vary because the route can be different from one flow to another for the same source-destination pair.

To facilitate modeling, we adopt a discrete time model where time is divided into consecutive time slots $t = 1, 2, \dots$, and the t -th flow arrives at the beginning of time slot t . It is assumed, without loss of generality, that flow requests arrive sequentially. For a flow f arriving at time t , we note $U(t)$ the utility function of the t -th flow. The objective is then:

$$\text{maximize } \lim_{T \rightarrow \infty} \sum_{t=1}^T U(t) \quad (4.1)$$

For this purpose, we propose an FDRL agent composed of an FL aggregator (i.e., the root controller) and FDRL learning nodes (i.e., the SDN controllers). Before presenting the design of the FDRL agent, we provide a description of the DRL part.

2.3 Description of the DRL part

Here, we present the DRL mechanism that finds a path for a flow f from α to β while optimizing the QoS and energy efficiency parameters as described earlier.

The state space is designed to capture the necessary information about the current state of the network. Therefore, we capture delay, loss, and throughput (all of which are QoS metrics). Hence, energy performance is evaluated based on the number of hops that make up the decided path.

The action space consists of all edges (i.e., links) of the network. The action space is defined as $A = \{a_1, a_2, \dots, a_{|E|}\}$, where each action corresponds to a link in the network $(i, j) \in E$. Actions are taken by the agent based on the observed state and in order to maximize the long-term cumulative reward.

The reward function is what directly affects the learning of the DRL algorithm. In our proposal, the reward function includes signals to deal with the optimization of the multiple metrics (i.e., maximizing the utility function), but also invalid actions and loops that may occur because the agent can decide any action at any time. Let $p \in \mathbb{R}_-$ be a penalty value. It should be set in a way to have the best balance between exploration of new paths and exploitation of previous knowledge. The reward function is defined as:

$$r(t) = \begin{cases} p & \text{if invalid} \\ U(t) & \text{else} \end{cases} \quad (4.2)$$

The agent's goal is to accumulate the maximum positive reward in each episode in order to have the maximum long-term cumulative reward. The utility function is defined as follows:

$$U(t) = w_1x_t - w_2d_t - w_3l_t - w_4h_t \quad (4.3)$$

where x_t is the throughput to maximize, d_t is the delay to minimize, l_t is the loss to minimize (i.e., maximize PDR), and h_t is the number of hops to minimize. All these metrics are weighted by $w_i \in \mathbb{R}_+, i \in \{1, 2, 3, 4\}$, which can be tuned to prioritize any metric during link selection.

We use the defined state space, action space, and reward function for DRL. To take an action, DRL uses DNN to approximate the reward that will result from the action. It outperforms RL in terms of handling larger state and action spaces. Our DNN takes the state space as input and returns a vector of action values as output.

We undertake a training process to enable our DRL agent to learn the optimal policy that maximizes the long-term cumulative reward. During this phase, our DRL algorithm iteratively updates the model parameters, represented as θ , in search of an improved policy. The end result of this training phase is the determination of the target model weights θ^* , which represent the optimal parameter configuration for our agent's policy.

2.4 Design of the FDRL routing

We build on the existing work on performing DRL routing in softwarized networking environments [58] [59] [60], and on the description of the DRL part presented earlier, to move towards a federated approach, namely FDRL routing. Figure 4.3 shows the behavior of our novel routing solution during the training phase.

As illustrated in the Gantt chart, each learning node, which corresponds to an SDN controller, trains its DRL agent locally, thus performing local weight updates $\theta^e, e \in N_e$, for a number of episodes N_e . After each FL round $r \in R$, the local model weights are sent to the FL aggregator, which is the root controller, which performs aggregation and sends back the updated global model weights. At the end of training, all nodes will have the final global model weights θ^* and will be able to have an optimized routing policy that ensures QoS performance and energy efficiency.

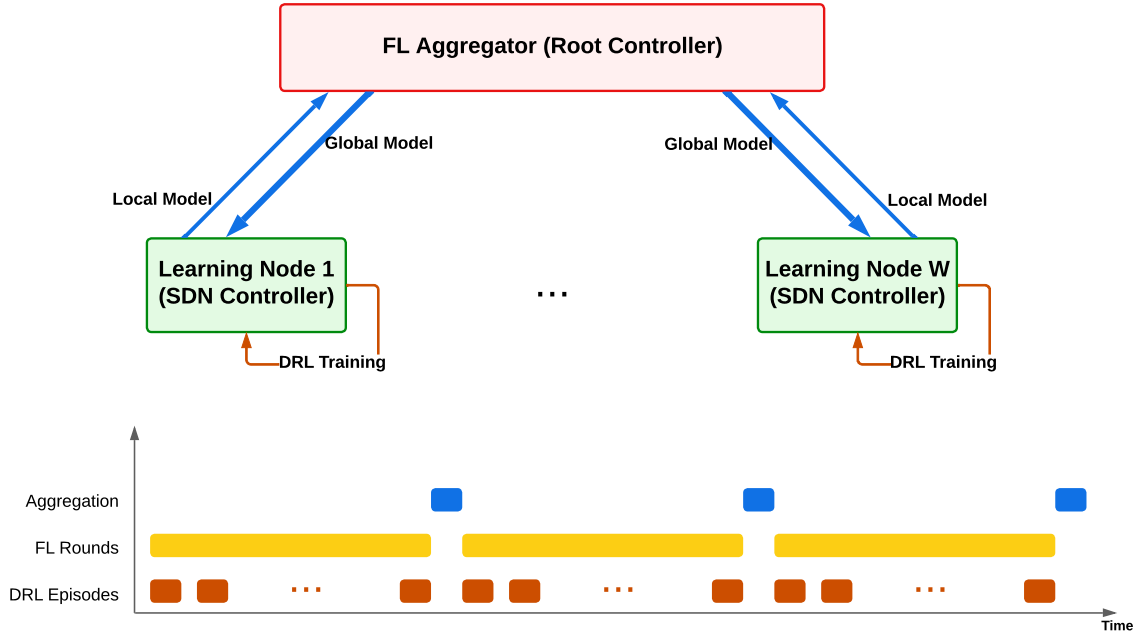


Figure 4.3: FDRL Training Phase

The Algorithm 1 is proposed to perform the FDRL aggregation. We use Federated Averaging (FedAvg) [61] to perform the aggregation and obtain the global model weights θ^r . Initial weights θ^0 are sent by the aggregator, and for each round r , local model weights are collected from the learning nodes, aggregated, and sent back as updated global model weights, and this is repeated until the end of the training. The final global model weights θ^* are sent to all learning nodes.

Algorithm 1: FDRL Aggregator Training Algorithm

Inputs : R : Number of rounds W : Number of learning nodes**Output:** θ^* : Global model weights

```
1 send initial weights  $\theta^0$  to learning nodes
2 for  $r \leftarrow 1$  to  $R$  do
3   for  $w \leftarrow 1$  to  $W$  do
4     wait for  $\theta^{w,r}$ 
5    $\theta^r \leftarrow \frac{1}{W} \sum_{w=1}^W \theta^{w,r}$ 
6   send updated weights  $\theta^r$  to learning nodes
7  $\theta^* \leftarrow \theta^r$ 
```

We now propose Algorithm 2, which permits the FDRL learning nodes training. Local DRL agents will perform N_e training episodes, each with a duration T . Given the current state s_t , the agent will decide and take an action a_t to get to state s_{t+1} and get immediate reward r_{t+1} .

Algorithm 2: FDRL Learning Node Training Algorithm

Inputs : A_g : Aggregation period N_e : Number of training episodes T : Episode horizon time ϵ : Exploration parameter**Output:** θ^* : Global model weights

```
1 wait for initial global weights  $\theta^g$  from aggregator
2 for  $e \leftarrow 1$  to  $N_e$  do
3    $\theta^e \leftarrow \theta^g$ 
4   get current state  $s_t$ 
5   for  $t \leftarrow 1$  to  $T$  do
6     generate a random number  $r \in [0, 1]$ 
7      $a_t \leftarrow \begin{cases} \text{pick an action that maximizes the reward} & \text{if } r > \epsilon \\ \text{pick a random action } a \in A & \text{else} \end{cases}$ 
8     take the action  $a_t$  to get to state  $s_{t+1}$  and get reward  $r_{t+1}$ 
9     store  $\{s_t, a_t, s_{t+1}, r_{t+1}\}$ 
10  update  $\theta^e$ 
11  if  $e \bmod A_g = 0$  then
12    send  $\theta^e$  to the aggregator
13    wait for the aggregated value  $\theta^g$ 
14  $\theta^* \leftarrow \theta^g$ 
```

Note that the exploration parameter ϵ impacts the balance between exploring new possibilities and exploiting the current best-known actions. When the time comes for aggregation, the local model weights are sent to the aggregator, which sends back the updated global model weights θ^g . At the end of the training process, each learning node will have the final global model weights θ^* .

3 Enhancing routing in softwarized IoT with NFV

Here, we propose to take advantage of the proposed general architecture of intelligent distributed network softwarization for IoT to further extend the proposal of a novel routing mechanism for softwarized IoT, thanks to NFV.

Some recent proposals have been made by academia to use DRL to perform service provisioning in NFV-enabled networks [53]. We propose to extend this proposal to intelligently place routing VNFs in a dynamic IoT network. We motivate our proposal by the fact that in an IoT network where constrained devices are enabled to act as routers, it is interesting to enable or disable this feature depending on QoS metrics and available power on the devices (since routing is an energy consuming task). The global view is already provided by the aforementioned integration of intelligent distributed SDN in IoT networks.

As depicted in Figure 4.4, the dynamic placement of routing functions will be ensured by the NFV MANO layer, to enhance the routing. It is highly motivated to use intelligent approaches for the placement.

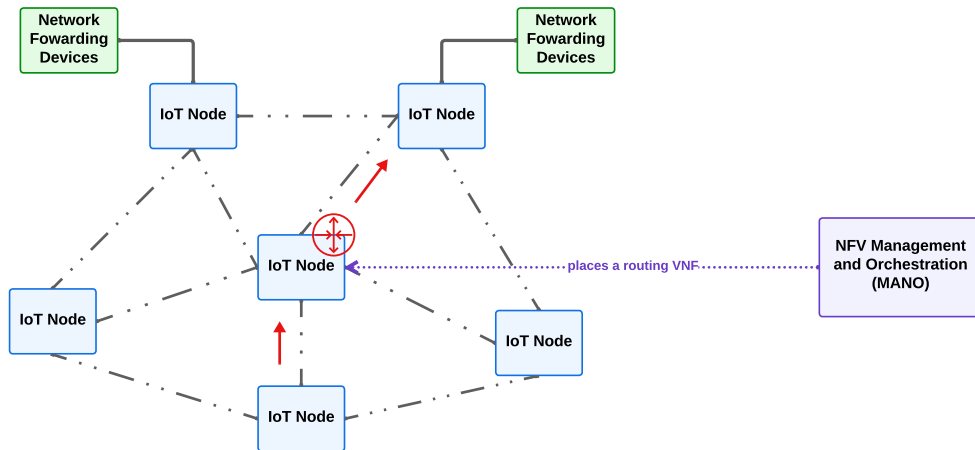


Figure 4.4: Improving Routing with NFV

In the example shown, the NFV MANO decides to place a routing VNF in a specific IoT node to enable communication between all nodes and the static network infrastructure (i.e., network forwarding devices). The choice of where to place the VNF is critical to improve QoS metrics and network lifetime (by saving energy on the devices).

Conclusion

In this chapter, we proposed a general architecture for intelligent distributed network softwarization for IoT based on the strengths of SDN, NFV, and ML. It has been taken as a starting point to propose a detailed novel routing algorithm for IoT softwarized networks, based on FDRL. Finally, we proposed to enhance the routing by using intelligent NFV techniques. Next, we will perform simulations and evaluations to establish the effectiveness of our contribution.

5

Simulation and Performance Evaluation

Introduction

In this final chapter, we demonstrate the effectiveness of our solution through simulation and performance evaluation. Considering the complexity of the proposal and the fact that it contains many novelties, we adopt a divide and conquer evaluation approach, where we test each novelty (i.e., sub-solution) separately. This allows us to easily interpret the results by being able to identify the cause behind each effect. After introducing the simulation environment and the performance metrics to be observed (e.g., delay, throughput), we present the experimental results and discuss the impact of distributed SDN control and intelligent routing on the softwarized IoT network. We also justify the overall effectiveness of our solution and present the future work that needs to be done to achieve a fully effective softwarization of IoT networks.

1 Simulation environment

In this section, we introduce the hardware and software simulation environment. In particular, we focus on explaining the choice of the network simulation tool and the SDN controller. As for the hardware environment, the simulations are performed on a computer equipped with an AMD Ryzen 7 4000 series processor supported by 16 GB of memory. The computer runs on Ubuntu 22.0.4, a Linux distribution.

We have chosen Python as the main programming and scripting language for our project. Python [62] is a general-purpose, high-level programming language. It is dynamically typed and interpreted (i.e., code is not converted to machine code before execution). Its simplicity and readability make it an ideal choice for rapid prototyping. Python is a popular language for many different tasks, including: software development, ML, automation, and networking, thanks to its extensive standard library and a vast ecosystem of third-party packages. In our case, it is well suited as it is fully compatible with the selected network simulation tool and SDN controller.

In order to carry out the network simulation, since no physical testbeds are available, we proceed with Mininet, which is well positioned compared to its counterparts, as evidenced by the comparison proposed in Table 5.1. Mininet is a network emulator that facilitates the creation of network topologies for research purposes. It allows the emulation of networks with virtual hosts, switches, and controllers, all running on real Linux kernels. It makes it easy to prototype and experiment various scenarios. We recall that network emulation provides better and more accurate results than simulation because it replicates real-world behavior by running the network software on virtualized hosts [57].

Table 5.1: Comparison of NS3, OMNeT++, and Mininet

	NS3 [63]	OMNeT++ [64]	Mininet [65]
Type	Simulator	Simulator	Emulator
Compatibility	Cross-platform	Cross-platform	Linux-based
Real-time emulation	Good	Good	Excellent
SDN integration	Supported	Supported	Smooth integration
Language	C++	C++	Python

For the SDN controller, we opt for the Ryu SDN controller, which is one of the most powerful controllers, as shown in the comparison in Table 5.2, and the choice is strongly driven by the fact that there is a strong complementarity between Mininet and Ryu. Ryu is an open source SDN controller that offers extensive programmability, which has made it a popular choice for network research. It provides an easily extensible environment for building SDN applications, implementing custom network logic, and integrating different network protocols, all via its libraries and RESTful API.

Table 5.2: Comparison of Floodlight, POX, and Ryu

	Floodlight [66]	POX [67]	Ryu [68]
Ease of setup	Moderate	Easy	Easy
Northbound API	RESTful API and libraries	Libraries	RESTful API and libraries
Documentation	Limited	Good	Good
Extensibility	Less extensible	Highly extensible	Highly extensible
Language	Java	Python	Python

We have presented an overview of the simulation environment, which is built with regard to the complementarity between the different tools. As an Integrated Development Environment (IDE), we have chosen PyCharm Community Edition, a robust and feature-rich development environment tailored for Python programming [69].

2 Metrics presentation

Before introducing the results, we present the key metrics used in the evaluation. Hence, we present the delay, throughput, and loss ratio metrics, summarized in Table 5.3. Further evaluation may be based on other QoS or energy-related performance indicators.

Delay, or latency, is one of the most common and important QoS metrics. It measures the time it takes for information to travel from source to destination in a network. Low latency is critical for real-time applications, and high latency can result in a poor user experience. Delays can be caused by the distance between the communicating pair, the quality of the network equipment and operation, etc. Delay is typically expressed in milliseconds (ms).

Throughput is also one of the most common QoS metrics because it is required by many applications (e.g., security video streaming). It measures the rate at which data can be successfully sent or received over a network and reflects the effective capacity of the network link. It is affected by factors such as bandwidth, latency, packet loss, and network congestion. Throughput is typically expressed in bits per second (bps), kilobits per second (Kbps), megabits per second (Mbps), or even gigabits per second (Gbps), depending on the capabilities of the network.

Loss ratio is a relevant QoS metric, especially when some links are unreliable and the possibility of loss exists. It measures the proportion of data packets that are lost or dropped during transmission over the network. In scenarios where network reliability is critical, such as critical data exchanges, a low loss ratio is essential to ensure that data is delivered accurately.

Table 5.3: Summary of Considered Metrics

Metric	Description	Unit	Ideal value
Delay	Measures the time it takes for data to travel from source to destination.	ms	The lowest
Throughput	Measures the rate at which data can be successfully transmitted across a network.	Mbps	The highest
Loss ratio	Represents the proportion of data packets that are lost or dropped during transmission.	%	The lowest

Please note that for a more accurate evaluation, other performance metrics such as jitter can be considered. Jitter measures the variation in packet delay, which is critical for assessing the stability of real-time applications. Energy-related metrics should also be considered to assess energy efficiency (e.g., number of hops, network lifetime, etc.).

In the remainder of this chapter, we retain delay and throughput as the main evaluation metrics, and we consider the loss ratio only if it exists due to a random loss probability introduced in some links. Otherwise, the loss is assumed to be equal to zero for all test cases.

3 Results and discussion

After setting up the environment, we run simulations to determine the effectiveness of our proposal. As presented, we proceed in two main steps. First, we evaluate the effect of distributed SDN control on the performance of IoT networks. Then, we evaluate the effect of intelligent routing on the performance of softwarized IoT networks.

3.1 Effect of distributed control

Here, we perform simulations to infer the impact of a distributed SDN control architecture on network performance. We measure the delay and throughput metrics in the case of centralized control (i.e., one SDN controller) and in the case of distributed control (i.e., two or more SDN controllers).

We take the network topology illustrated in Figure 5.1, adopted with considerable changes from proposal [70]. It consists of 16 forwarding devices and 3 hosts $\{h1, h2, h3\}$. We propose 3 test cases: 4 SDN controllers $\{c1, c2, c3, c4\}$, each controlling a group of forwarding devices (as in Figure 5.1), 2 SDN controllers where $c1'$ controls the devices of $\{c1, c2\}$ and where $c2'$ controls the devices of $\{c3, c4\}$, and 1 SDN controller that controls all forwarding devices (i.e., the common SDN controller design). This topology is emulated via Mininet using a script we wrote for this purpose.

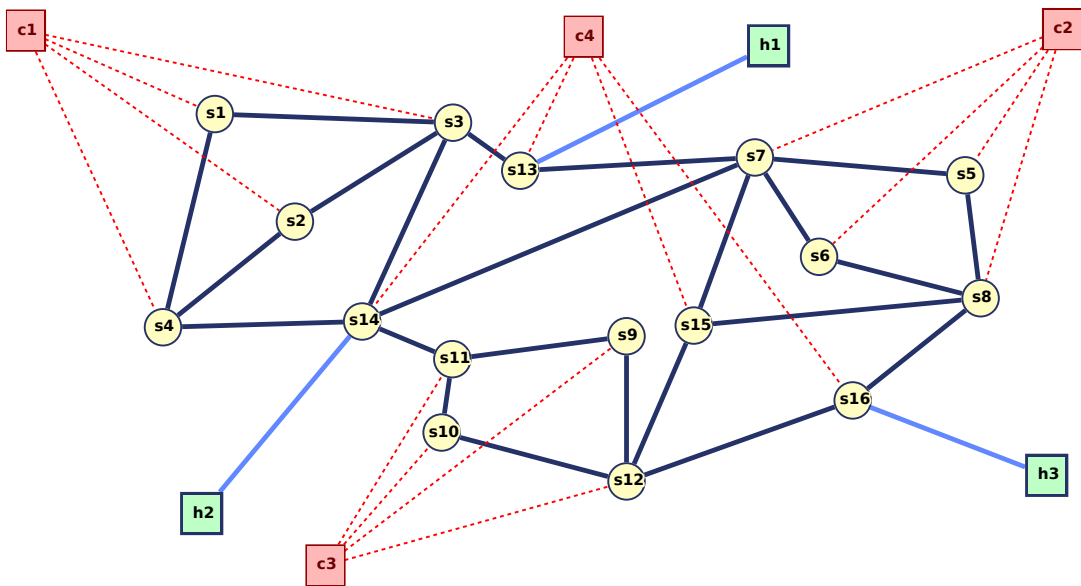


Figure 5.1: Adopted Network Topology to Evaluate Distributed Control

We consider SDN controllers that all use a basic routing strategy (i.e., SPR). We instantiate the Ryu controllers, run the Mininet script that creates and connects the hosts and switches, and connects the switches to the remote controllers. Using *ping*, a widely used network diagnostic and testing tool, we take 250 Round-Trip Time (RTT, or latency) measurements with 64 bytes of data to evaluate the delay when *h1* pings *h2*, when *h1* pings *h3*, and when *h2* pings *h3*. In Figure 5.2, we present a screenshot of the delay measurement setup.

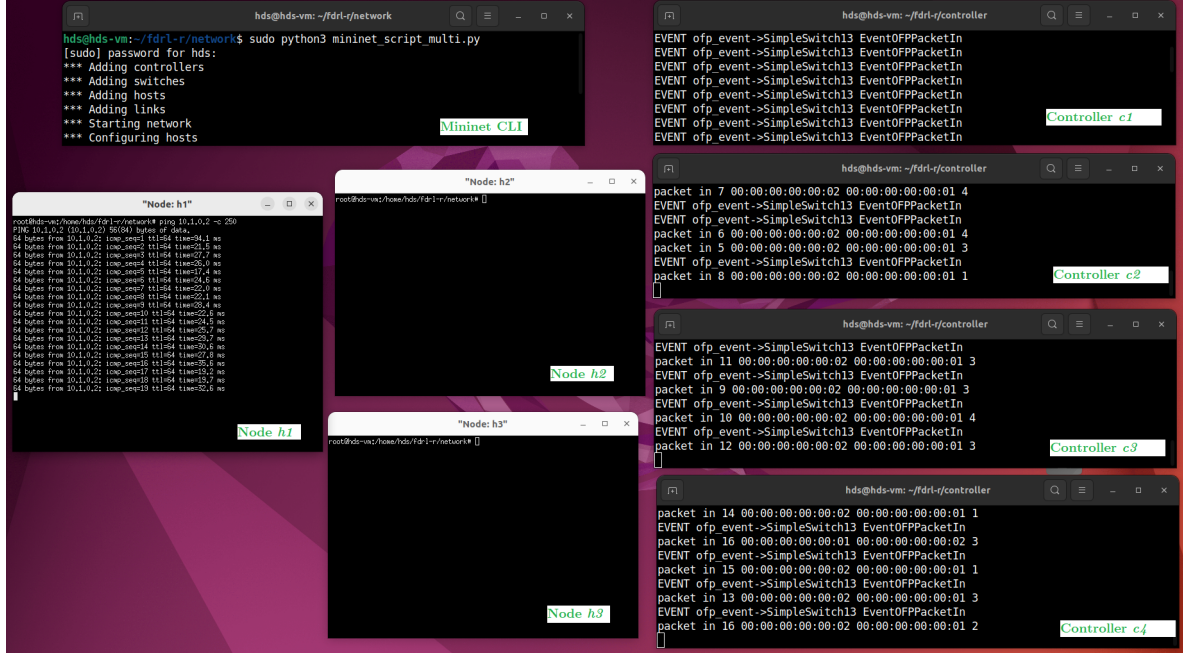


Figure 5.2: Screenshot of Delay Measurement Setup

We perform the measurements for three cases: with 1, 2, and 4 SDN controllers leading the network forwarding devices. The results are shown in Figure 5.3. For each source-destination pair, we plot the delay values for the three cases.

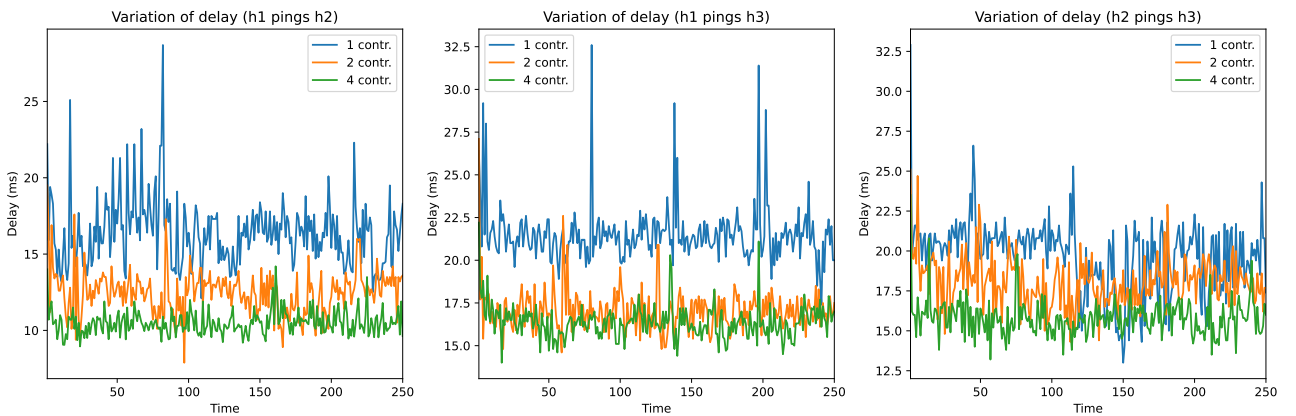


Figure 5.3: Variation of Delay with Time

We observe that for the three source-destination pairs, as we increase the number of SDN controllers, moving towards a distributed control design, the delay decreases for almost all the measures. Delay varies from one measure to another because it depends on the network state, but also due to the instability of the emulation environment (since the measures are performed in a virtual environment). For more accurate results, it is better to move to a testbed.

Before drawing conclusions, we suggest to also evaluate the average throughput between the same source-destination pairs and for the three test cases (i.e., 1, 2, or 4 controllers). For this purpose, we use the *iperf3* tool. It is a powerful network performance testing tool designed to measure the throughput, bandwidth, and quality of network connections. We perform a 100 second duration test, for three times. To test the throughput between $h1$ and $\{h2, h3\}$, the host $h1$ acts as *iperf3* server and the others as clients. Similarly, $h2$ acts as a server and $h3$ as a client to measure the throughput between $h2$ and $h3$. Figure 5.4 shows a screenshot of the throughput measurement setup.

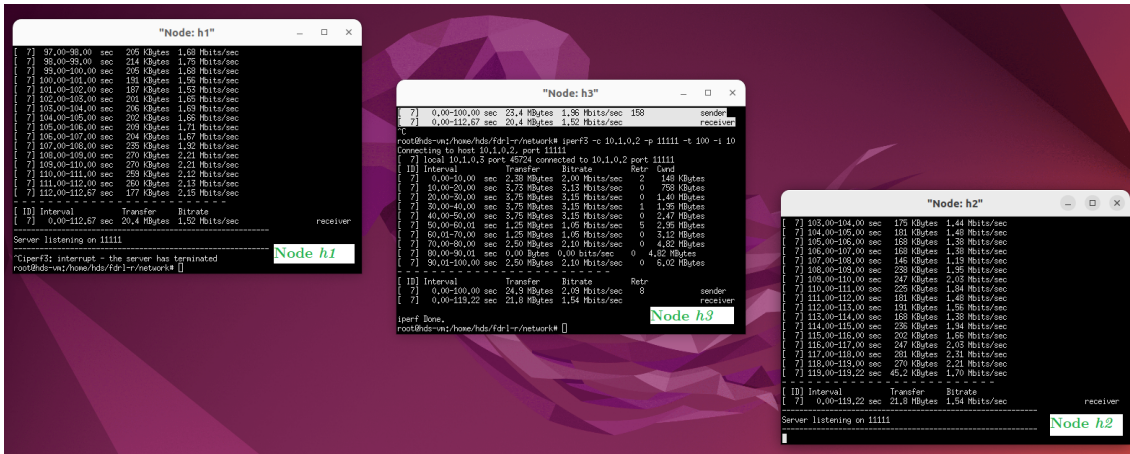


Figure 5.4: Screenshot of Throughput Measurement Setup

We perform the measurements and calculate the average throughput values for each case. In Figure 5.5, we plot the average delay and average throughput versus the number of controllers (1, 2, or 4).

We notice that the average delay observed between $h1$ and $h2$ is always lower than the other source-destination pairs. This is justified if we calculate the number of hops of the shortest path between the hosts using the Dijkstra algorithm: $h1$ and $h2$ are separated by 4 hops, while the other two pairs are separated by 6 hops. We do not observe this difference in the average throughput graph, which can be explained by the fact that there is no loss in the network and that all links of the topology have the same bandwidth.

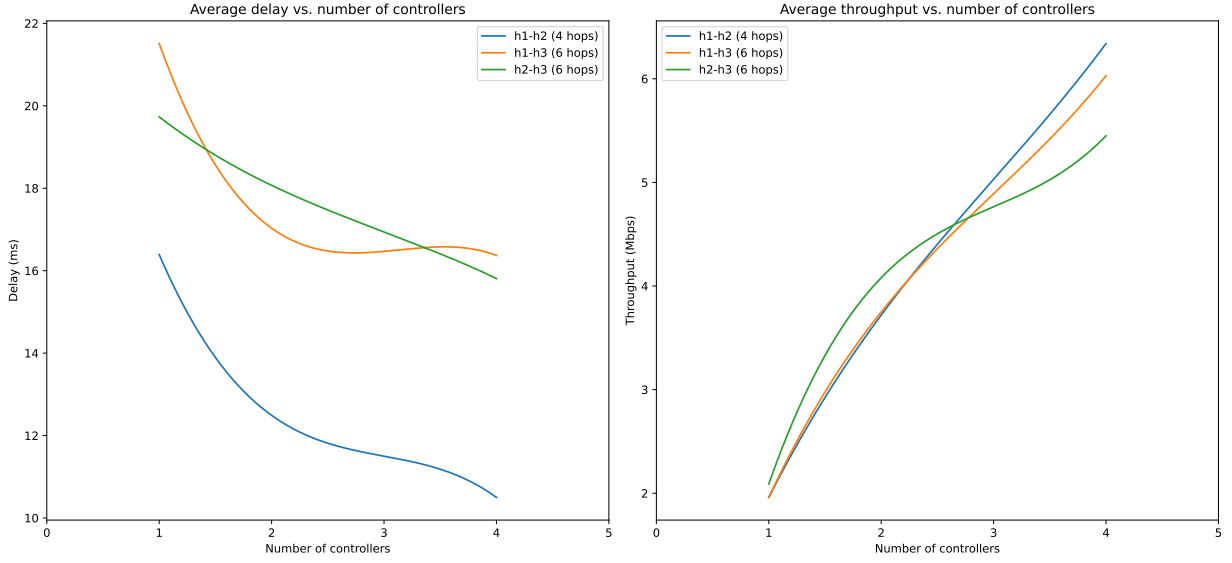


Figure 5.5: Average Delay and Throughput

As the number of SDN controllers increases, moving from a centralized to a distributed controller design, the delay decreases while the throughput increases, meaning that the overall network performance increases.

Based on these results and observations, we conclude that moving towards distributed control has a great potential to increase the overall network performance as shown by the evaluated quality metrics. One of the main reasons for this is that the distributed design reduces the load on the controllers, thus speeding up the processing time, especially for large-scale IoT networks.

3.2 Effect of intelligent routing

Now, we perform the necessary simulations to evaluate the outcome of having intelligence in the routing process in softwarized IoT networks. For that purpose, we consider the DRL-based routing method proposed by authors in [59].

We evaluate this intelligent routing approach in two common network topologies found in IoT scenarios: Abilene and GEANT. The reason for having two topologies is to challenge the DRL agent, since GEANT has more nodes and links than Abilene, and thus a larger input state space. Topologies are depicted in Figure 5.6. In the conducted evaluation, one controller is responsible of all the network. A 10% random packet loss is added to the network. DRL routing is evaluated in terms of delay, throughput, and loss ratio. Figure 5.7 shows a screenshot of the intelligent routing evaluation setup.

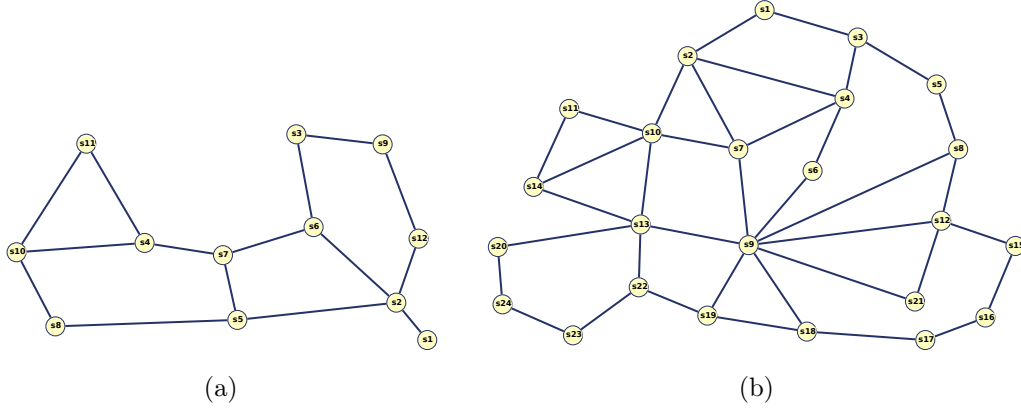


Figure 5.6: Adopted Network Topologies: Abilene (a) and GEANT (b)

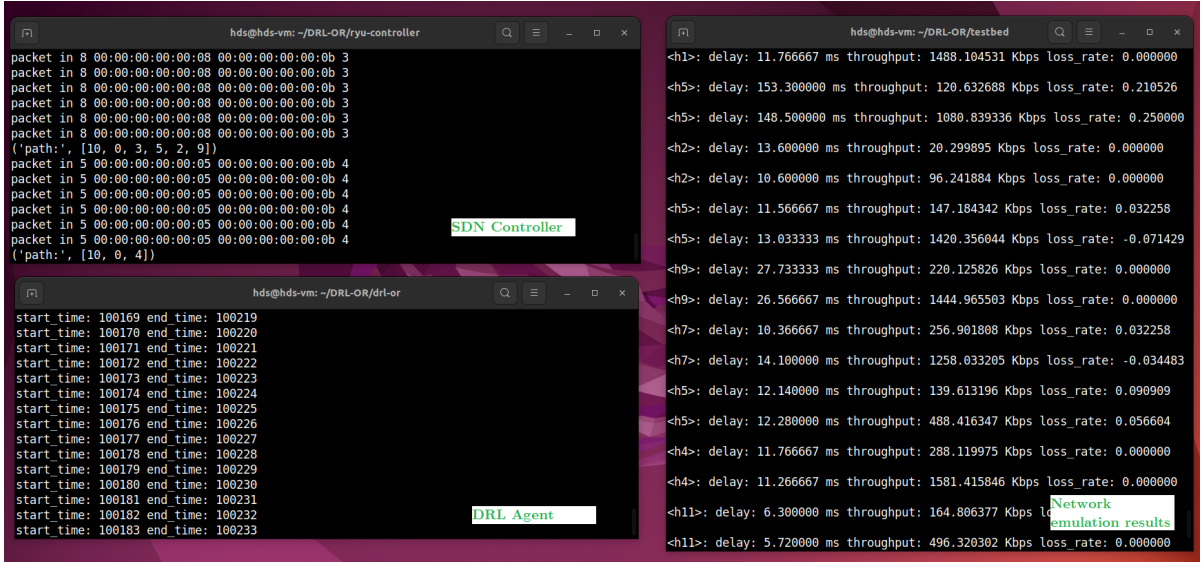


Figure 5.7: Screenshot of Intelligent Routing Evaluation Setup

Results are presented in Figure 5.8. We present the average latency, the average throughput ratio (calculated according to bandwidth), and the average loss ratio. For all scenarios and topologies, and for all three metrics, intelligent DRL-based routing outperforms conventional SPR. We also note that DRL routing still outperforms SPR in the GEANT topology, although it is more challenging because it is larger and causes a much larger input state space. We recall that this problem is one of the main reasons for moving from RL to DRL, since DNN allows us to handle larger state and action spaces.

Based on these results, we conclude on the real effectiveness of having intelligence in the network, especially for routing optimization, as it outperforms its counterparts in terms of improving performance metrics (lower delay, higher throughput ratio, and lower loss ratio).

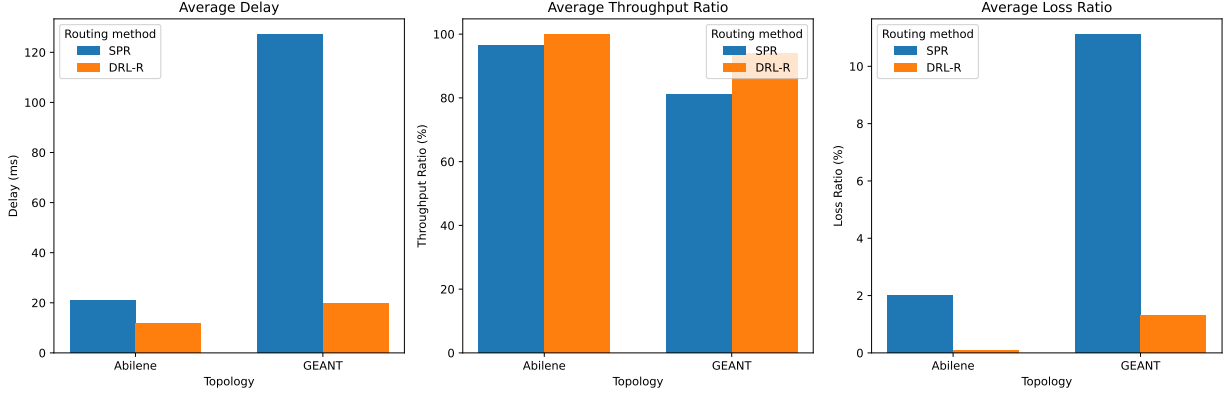


Figure 5.8: Intelligent Routing Evaluation Results

Considering the results of our two main experiments and the conclusions drawn, we finally conclude that the presence of distributed control and intelligence in the network greatly improves the routing, and thus improves the overall performance of the network. We have shown the effectiveness of our proposal to move towards intelligent distributed network softwarization for IoT, and to consider intelligent routing.

As future work, we propose to implement the complete solution of FDRL routing in softwarized IoT environment. In the present work, we have proven the effectiveness of distributed control and intelligence as two boosters for the performance of softwarized IoT networks. Since there are no conflicts or redundancies between these two novelties, we strongly support the effectiveness of the overall solution.

Conclusion

In this concluding chapter, we have performed the necessary simulations to evaluate the impact of our proposal on network performance. Extensive simulation has shown that having distributed control and intelligent routing significantly improves the overall network performance (e.g., lower latency and higher throughput). It has been demonstrated that our two novelties outperform classical SPR. Since there is no conflict between them, the conducted experiments demonstrate the effectiveness of the proposal to move towards intelligent distributed network softwarization for IoT and to consider intelligent FDRL-based routing. This is what we set as future work, as we need to confirm the results through the integral implementation of the routing solution.

Conclusion and Perspectives

It is undeniable that there is an urgent need to provide agile and scalable IoT networks by mitigating their many constraints, as IoT is now an integral part of everyday life. Quality and energy efficiency requirements need to be fully satisfied, while computational and power usage need to be optimized.

In this work, we have explored network softwarization (mainly, SDN and NFV) and ML with a vision to combine them with IoT, thus moving towards intelligent distributed network softwarization for IoT constrained networks.

After thoroughly investigating the main concepts of IoT, network softwarization, and ML, and paying special attention to their architectures, we have reviewed the state of the art in combining these three promising technologies. After surveying the most relevant and recent work on the topic, we concluded that work exploiting the potential of such a combination is still sparse.

Accordingly, our contribution has been to move towards intelligent distributed network softwarization for IoT. After proposing the general architecture, we have proposed to take advantage of it by proposing a new routing mechanism to optimize routing in softwarized IoT. This FDRL-based solution is supported by both distributed control and intelligent routing novelties. Then, we have formalized the routing problem and provided the detailed description and design of the FDRL routing. We have also highlighted how NFV can be a booster for our contribution.

Extensive simulation and performance evaluation of the novelties have shown the effectiveness of the proposal. Both distributed SDN control and intelligent routing have a significant positive impact on the investigated quality metrics (delay, throughput, and loss ratio) in a softwarized IoT network environment. Our novelties outperform the conventional SPR counterpart.

In summary, during this 6-month internship, we have achieved the investigation of the combination of IoT, network softwarization, and ML. For this purpose, we conducted a comprehensive literature review, and then proposed and designed our solution to move to an intelligent distributed network softwarization for IoT. We have conducted the necessary experiments, and the results confirm our findings. During this period, we have also made several contributions to conferences and journals, as evidenced earlier in this report.

As future work, we propose to develop a prototype of the full softwarization solution to demonstrate its positive outcome and to analyze the different performance metrics, having as perspective to ensure a real implementable solution. We also consider moving to performing evaluations in testbeds, for more accurate results. In order to do this speedily, it is necessary to have access to the necessary resources (e.g., network datasets, source codes). We strongly support having high quality normalized network datasets to perform coherent performance analysis. This requires the combined efforts of the entire research community, as this has been the major difficulty we have faced during the elaboration of the present work.

Finally, it is very interesting to examine the future directions that will be achieved thanks to the promising technologies of IoT, network softwarization, and ML. Our proposal of intelligent distributed network softwarization for IoT can be combined with more emerging technologies such as Edge Computing (EC) to provide shorter service response time. It is also crucial to consider the security aspect and use this novel architecture to protect IoT from potential security risks. In fact, this is the key to having real implementable solutions in industrial and manufacturing environments, and intelligence can lead towards a self-secured IoT.

Bibliography

- [1] Yanghao Xie, Lin Huang, Yuyang Kong, Sheng Wang, Shizhong Xu, Xiong Wang, and Jing Ren. Virtualized Network Function Forwarding Graph Placing in SDN and NFV-Enabled IoT Networks: A Graph Neural Network Assisted Deep Reinforcement Learning Method. *IEEE Trans. Netw. Serv. Manage.*, 19(1):524–537, March 2022.
- [2] Seyha Ros, Prohim Tam, and Seokhoon Kim. Modified Deep Reinforcement Learning Agent for Dynamic Resource Placement in IoT Network Slicing. *Journal of Internet Computing and Services*, 23(5):17–23, October 2022.
- [3] Elham Hajian, Mohammad Reza Khayyambashi, and Naser Movahhedinia. A Mechanism for Load Balancing Routing and Virtualization Based on SDWSN for IoT Applications. *IEEE Access*, 10:37457–37476, 2022.
- [4] Ali Haider Shamsan and Arman Rasool Faridi. A Novel SDNFV IoT Architecture Leveraging Softwarization Technology Services to Alleviate IoT Network Resource Restrictions. *IJETT*, 70(2):1–10, February 2022.
- [5] Rashid Amin, Elisa Rojas, Aqsa Aqdus, Sadia Ramzan, David Casillas-Perez, and Jose M. Arco. A Survey on Machine Learning Techniques for Routing Optimization in SDN. *IEEE Access*, 9:104582–104611, 2021.
- [6] Anteneh A. Gebremariam, Muhammad Usman, and Marwa Qaraqe. Applications of Artificial Intelligence and Machine Learning in the Area of SDN and NFV: A Survey. In *2019 16th International Multi-Conference on Systems, Signals & Devices (SSD)*, pages 545–549, Istanbul, Turkey, March 2019. IEEE.
- [7] Bassem Sellami, Akram Hakiri, Sadok Ben Yahia, and Pascal Berthou. Deep Reinforcement Learning for Energy-Efficient Task Scheduling in SDN-based IoT Network.

- In *2020 IEEE 19th International Symposium on Network Computing and Applications (NCA)*, pages 1–4, Cambridge, MA, USA, November 2020. IEEE.
- [8] UTC website. <https://www.utc.fr/en/>. Consulted on July 26, 2023.
- [9] CNRS website. <https://www.cnrs.fr/en/>. Consulted on July 26, 2023.
- [10] Heudiasyc website. <https://www.hds.utc.fr/en/>. Consulted on July 26, 2023.
- [11] Heudiasyc. Rapport d’activités 2022 UMR UTC-CNRS 7253, 2023. Internal diffusion.
- [12] Abbas Yazdinejad, Reza M. Parizi, Ali Dehghantanha, Qi Zhang, and Kim-Kwang Raymond Choo. An Energy-Efficient SDN Controller Architecture for IoT Networks With Blockchain-Based Security. *IEEE Trans. Serv. Comput.*, 13(4):625–638, July 2020.
- [13] IoT Analytics. State of IoT 2023: Number of connected IoT devices. <https://iot-analytics.com/number-connected-iot-devices/>. Consulted on July 26, 2023.
- [14] University of Sydney. Creating research innovations and breakthroughs by thinking differently. <https://www.sydney.edu.au/research/research-impact/creating-research-innovations-breakthroughs-by-thinking-differently.html>. Consulted on July 26, 2023.
- [15] Ali Haider Shamsan and Arman Rasool Faridi. Network softwarization for IoT: A Survey. 2019.
- [16] IoT Solutions World Congress. 10 cutting-edge IoT use cases for 2023. <https://www.iotsworldcongress.com/10-cutting-edge-iot-use-cases-for-2023/>. Consulted on July 27, 2023.
- [17] C. C. Sobin. A Survey on Architecture, Protocols and Challenges in IoT. *Wireless Pers Commun*, 112(3):1383–1429, June 2020.
- [18] Mohamed S. Abdalzaher, Hussein A. Elsayed, Mostafa M. Fouda, and Mahmoud M. Salim. Employing Machine Learning and IoT for Earthquake Early Warning System in Smart Cities. *Energies*, 16(1):495, January 2023.
- [19] Sarah A. Al-Qaseemi, Hajer A. Almulhim, Maria F. Almulhim, and Saqib Rasool Chaudhry. IoT architecture challenges and issues: Lack of standardization. In *2016*

- Future Technologies Conference (FTC)*, pages 731–738, San Francisco, CA, USA, December 2016. IEEE.
- [20] Seyyed Keyvan Mousavi, Ali Ghaffari, Sina Besharat, and Hamed Afshari. Security of internet of things based on cryptographic algorithms: a survey. *Wireless Netw*, 27(2):1515–1555, February 2021.
- [21] Laith Farhan, Sinan T. Shukur, Ali E. Alissa, Mohmad Alrweg, Umar Raza, and Rupak Kharel. A survey on the challenges and opportunities of the Internet of Things (IoT). In *2017 Eleventh International Conference on Sensing Technology (ICST)*, pages 1–5, Sydney, NSW, December 2017. IEEE.
- [22] Bo Yi, Xingwei Wang, Keqin Li, Sajal K. Das, and Min Huang. A comprehensive survey of Network Function Virtualization. *Computer Networks*, 133:212–262, March 2018.
- [23] Adrian Popescu and Alexander Westerhagen. Network Softwarization: Developments and Challenges. In *2022 14th International Conference on Communications (COMM)*, pages 1–6, Bucharest, Romania, June 2022. IEEE.
- [24] Ali Haider Shamsan and Arman Rasool Faridi. SDN-Assisted IoT Architecture: A Review. In *2018 4th International Conference on Computing Communication and Automation (ICCCA)*, pages 1–7, Greater Noida, India, December 2018. IEEE.
- [25] Abdallah Ouhab, Thiago Abreu, Hachem Slimani, and Abdelhamid Mellouk. Energy-efficient clustering and routing algorithm for large-scale SDN-based IoT monitoring. In *ICC 2020 - 2020 IEEE International Conference on Communications (ICC)*, pages 1–6, Dublin, Ireland, June 2020. IEEE.
- [26] Faiza Mahmood Shuker. Improvement of Routing Protocol for IoT Network Using SDN/OpenFlow. In *2019 1st AL-Noor International Conference for Science and Technology (NICST)*, pages 54–57, Sulimanyiah - Kurdistan Region- IRAQ, Iraq, October 2019. IEEE.
- [27] Junfeng Xie, F. Richard Yu, Tao Huang, Renchao Xie, Jiang Liu, Chenmeng Wang, and Yunjie Liu. A Survey of Machine Learning Techniques Applied to Software Defined Networking (SDN): Research Issues and Challenges. *IEEE Commun. Surv. Tutorials*, 21(1):393–430, 2019.

- [28] Rashid Mijumbi, Joan Serrat, Juan-Luis Gorricho, Niels Bouten, Filip De Turck, and Raouf Boutaba. Network Function Virtualization: State-of-the-Art and Research Challenges. *IEEE Commun. Surv. Tutorials*, 18(1):236–262, 2016.
- [29] Surendra Kumar Keshari, Vineet Kansal, Sumit Kumar, and Priti Bansal. An intelligent energy efficient optimized approach to control the traffic flow in Software-Defined IoT networks. *Sustainable Energy Technologies and Assessments*, 55:102952, February 2023.
- [30] Michael Baddeley, Reza Nejabati, George Oikonomou, Mahesh Sooriyabandara, and Dimitra Simeonidou. Evolving SDN for Low-Power IoT Networks. In *2018 4th IEEE Conference on Network Softwarization and Workshops (NetSoft)*, pages 71–79, Montreal, QC, June 2018. IEEE.
- [31] Nikos Bizanis and Fernando A. Kuipers. SDN and Virtualization Solutions for the Internet of Things: A Survey. *IEEE Access*, 4:5591–5606, 2016.
- [32] Alireza Shirmarz and Ali Ghaffari. Performance issues and solutions in SDN-based data center: a survey. *J Supercomput*, 76(10):7545–7593, October 2020.
- [33] Juliver Gil Herrera and Juan Felipe Botero. Resource Allocation in NFV: A Comprehensive Survey. *IEEE Trans. Netw. Serv. Manage.*, 13(3):518–532, September 2016.
- [34] Abdelquoddouss Laghrissi and Tarik Taleb. A Survey on the Placement of Virtual Resources and Virtual Network Functions. *IEEE Commun. Surv. Tutorials*, 21(2):1409–1434, 2019.
- [35] Vasileios A. Memos and Konstantinos E. Psannis. NFV-Based Scheme for Effective Protection against Bot Attacks in AI-Enabled IoT. *IEEE Internet Things M.*, 5(1):91–95, March 2022.
- [36] Hye-Jin Ku, Joo-hee Jung, and Gu-In Kwon. A Study on Reinforcement Learning based SFC Path Selection in SDN/NFV. 12(12), 2017.
- [37] Elia Guerra, Francesc Wilhelmi, Marco Miozzo, and Paolo Dini. The Cost of Training Machine Learning Models over Distributed Data Sources. *IEEE Open J. Commun. Soc.*, pages 1–1, 2023.

- [38] Jingjing Zheng, Kai Li, Naram Mhaisen, Wei Ni, Eduardo Tovar, and Mohsen Guizani. Federated Learning for Online Resource Allocation in Mobile Edge Computing: A Deep Reinforcement Learning Approach. In *2023 IEEE Wireless Communications and Networking Conference (WCNC)*, pages 1–6, Glasgow, United Kingdom, March 2023. IEEE.
- [39] Dinh C. Nguyen, Ming Ding, Pubudu N. Pathirana, Aruna Seneviratne, Jun Li, and H. Vincent Poor. Federated Learning for Internet of Things: A Comprehensive Survey. *IEEE Commun. Surv. Tutorials*, 23(3):1622–1658, 2021.
- [40] Taotao Zhao, Xiaomin Chen, Qiang Sun, and Jiayi Zhang. Energy-Efficient Federated Learning Over Cell-Free IoT Networks: Modeling and Optimization. *IEEE Internet Things J.*, pages 1–1, 2023.
- [41] Latif U. Khan, Walid Saad, Zhu Han, Ekram Hossain, and Choong Seon Hong. Federated Learning for Internet of Things: Recent Advances, Taxonomy, and Open Challenges. *IEEE Commun. Surv. Tutorials*, 23(3):1759–1799, 2021.
- [42] Ahmed Imteaj, Urmish Thakker, Shiqiang Wang, Jian Li, and M. Hadi Amini. A Survey on Federated Learning for Resource-Constrained IoT Devices. *IEEE Internet Things J.*, 9(1):1–24, January 2022.
- [43] Md Ferdous Pervej, Richeng Jin, and Huaiyu Dai. Resource Constrained Vehicular Edge Federated Learning with Highly Mobile Connected Vehicles. *IEEE J. Select. Areas Commun.*, pages 1–1, 2023.
- [44] Jiaju Qi, Qihao Zhou, Lei Lei, and Kan Zheng. Federated Reinforcement Learning: Techniques, Applications, and Open Challenges. *IR*, 2021. arXiv:2108.11887 [cs].
- [45] Wolfgang Kellerer, Patrick Kalmbach, Andreas Blenk, Arsany Basta, Martin Reisslein, and Stefan Schmid. Adaptable and Data-Driven Softwarized Networks: Review, Opportunities, and Challenges. *Proc. IEEE*, 107(4):711–731, April 2019.
- [46] Ricardo S. Alonso, Javier Prieto, Fernando de La Prieta, Sara Rodriguez-Gonzalez, and Juan M. Corchado. A Review on Deep Reinforcement Learning for the management of SDN and NFV in Edge-IoT. In *2021 IEEE Globecom Workshops (GC Wkshps)*, pages 1–6, Madrid, Spain, December 2021. IEEE.

- [47] Muhammad Aneeq Abid, Naokhaiz Afaqui, Muazzam A. Khan, Muhammad Waseem Akhtar, Asad Waqar Malik, Arslan Munir, Jawad Ahmad, and Balawal Shabir. Evolution towards Smart and Software-Defined Internet of Things. *AI*, 3(1):100–123, February 2022.
- [48] Dipon Saha, Meysam Shojaee, Michael Baddeley, and Israat Haque. An Energy-Aware SDN/NFV Architecture for the Internet of Things. 2020.
- [49] Alejandro Molina Zarca, Jorge Bernal Bernabe, Antonio Skarmeta, and Jose M. Alcaraz Calero. Virtual IoT HoneyNets to Mitigate Cyberattacks in SDN/NFV-Enabled IoT Networks. *IEEE J. Select. Areas Commun.*, 38(6):1262–1277, June 2020.
- [50] Alejandro Molina Zarca, Miloud Bagaa, Jorge Bernal Bernabe, Tarik Taleb, and Antonio F. Skarmeta. Semantic-Aware Security Orchestration in SDN/NFV-Enabled IoT Systems. *Sensors*, 20(13):3622, June 2020.
- [51] Israat Haque and Dipon Saha. SoftIoT: A resource-aware SDN/NFV-based IoT network. *Journal of Network and Computer Applications*, 193:103208, November 2021.
- [52] Hatim Jaadouni, Chaimae Saadi, and Habiba Chaoui. SDN/NFV architectures for edge-cloud oriented IoT. *ITM Web Conf.*, 46:02004, 2022.
- [53] Ali Nouruzi, Abolfazl Zakeri, Mohammad Reza Javan, Nader Mokari, Rasheed Hussain, and S. M. Ahsan Kazmi. Online Service Provisioning in NFV-Enabled Networks Using Deep Reinforcement Learning. *IEEE Trans. Netw. Serv. Manage.*, 19(3):3276–3289, September 2022.
- [54] Raheleh Samadi and Joeheon Seitz. Machine Learning Routing Protocol in Mobile IoT based on Software-Defined Networking. In *2022 IEEE Conference on Network Function Virtualization and Software Defined Networks (NFV-SDN)*, pages 108–111, Phoenix, AZ, USA, November 2022. IEEE.
- [55] Prohim Tam, Sa Math, Ahyoung Lee, and Seokhoon Kim. Multi-Agent Deep Q-Networks for Efficient Edge Federated Learning Communications in Software-Defined IoT. *Computers, Materials & Continua*, 71(2):3319–3335, 2022.

- [56] Guilherme Werneck De Oliveira, Michele Nogueira, Aldri Luiz Dos Santos, and Daniel Macêdo Batista. Intelligent VNF Placement to Mitigate DDoS Attacks on Industrial IoT. *IEEE Trans. Netw. Serv. Manage.*, pages 1–1, 2023.
- [57] N D Patel, B M Mehtre, and Rajeev Wankar. Simulators, Emulators, and Test-beds for Internet of Things: A Comparison. In *2019 Third International conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC)*, pages 139–145, Palladam, India, December 2019. IEEE.
- [58] Syed Qaisar Jalil, Mubashir Husain Rehmani, and Stephan Chalup. DQR: Deep Q-Routing in Software Defined Networks. In *2020 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8, Glasgow, United Kingdom, July 2020. IEEE.
- [59] Chenyi Liu, Mingwei Xu, Yuan Yang, and Nan Geng. DRL-OR: Deep Reinforcement Learning-based Online Routing for Multi-type Service Requirements. In *IEEE INFOCOM 2021 - IEEE Conference on Computer Communications*, pages 1–10, Vancouver, BC, Canada, May 2021. IEEE.
- [60] Peyman Tehrani, Francesco Restuccia, and Marco Levorato. Federated Deep Reinforcement Learning for the Distributed Control of NextG Wireless Networks. In *2021 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, pages 248–253, Los Angeles, CA, USA, December 2021. IEEE.
- [61] Adrian Nilsson, Simon Smith, Gregor Ulm, Emil Gustavsson, and Mats Jirstrand. A Performance Evaluation of Federated Learning Algorithms. In *Proceedings of the Second Workshop on Distributed Infrastructures for Deep Learning*, pages 1–8, Rennes France, December 2018. ACM.
- [62] Python 3.11.5 documentation. <https://docs.python.org/3/>. Consulted on September 3, 2023.
- [63] NS3 website. <https://www.nsnam.org/>. Consulted on September 4, 2023.
- [64] OMNeT++ website. <https://omnetpp.org/>. Consulted on September 4, 2023.
- [65] Mininet website. <https://mininet.org/>. Consulted on September 4, 2023.
- [66] Floodlight controller documentation. <https://floodlight.atlassian.net/wiki/spaces/floodlightcontroller/overview>. Consulted on September 4, 2023.

- [67] POX controller documentation. <https://noxrepo.github.io/pox-doc/html/>. Consulted on September 4, 2023.
- [68] Ryu controller documentation. <https://ryu.readthedocs.io/en/latest/>. Consulted on September 4, 2023.
- [69] PyCharm website. <https://www.jetbrains.com/pycharm/>. Consulted on September 4, 2023.
- [70] Shortest Path Routing. <https://github.com/amitsk1994/mininet-RYU-ShortestPath/>. Consulted on September 5, 2023.

تنمو إنترنت الأشياء (IoT) بشكل هائل، بل وصارت جزءاً أساسياً من الحياة اليوميّة. وتتكون إنترنت الأشياء من عدد هائل من الأجهزة المتنوعة والمقيدة، المتصلة ببعضها البعض، ما يطرح تحديات عديدة. من خلال هذا العمل، نقترح دراسة توظيف مجموعة من التكنولوجيات الحديثة كبرمجة الشبكات والتعلم الآلي للتغلب على التحديات وتحسين أداء شبكات إنترنت الأشياء. بعد إجراء دراسة معمقة للأعمال السابقة في المجال، نقترح بنية جديدة، ذكيّة وموزعة، لبرمجة شبكات إنترنت الأشياء، وتوظيفها لبناء آلية توجيه جديدة، تتسم بالذكاء. بناءً على المحاكاة والتجارب اللازمة، تمكنا من إبراز فاعلية مقترحنا.

الكلمات المفتاحية: إنترنت الأشياء (IoT)، برمجة الشبكات، شبكات معرفة بالبرمجيات (SDN)، التمثيل الافتراضي لوظائف الشبكات (NFV)، التعلم الآلي (ML).

Softwarisation Intelligente et Distribuée de Réseau pour Pallier aux Contraintes de l'Internet des Objets

Résumé

L'Internet des objets (IoT) est en pleine croissance et fait désormais partie intégrante de la vie quotidienne. Il est composé d'une quantité considérable de dispositifs interconnectés, hétérogènes et avec des ressources limitées, ce qui pose d'énormes défis. Dans ce travail, nous proposons d'étudier la combinaison des nouvelles technologies telles que la softwarisation du réseau et l'apprentissage automatique pour relever ces défis et améliorer les performances des réseaux IoT. Après un examen de l'état de l'art, une nouvelle architecture de softwarisation intelligente et distribuée des réseaux pour l'IoT est proposée et utilisée pour fournir un nouveau mécanisme de routage intelligent. Les simulations réalisées ont mis en évidence l'efficacité de notre solution.

Mots-clés : Internet des Objets (IoT), Softwarisation de Réseau, Réseaux Définis par Logiciel (SDN), Virtualisation des Fonctions Réseau (NFV), Apprentissage Automatique (ML).

Intelligent Distributed Network Softwarization to Alleviate Internet of Things Constraints

Abstract

The Internet of Things (IoT) is growing exponentially and has become an integral part of everyday life. It consists of a huge volume of interconnected heterogeneous and constrained devices, which poses enormous challenges. In this work, we propose to investigate the combination of emerging technologies such as network softwarization and machine learning to overcome the challenges and improve the performance of IoT networks. After a comprehensive review of the state of the art, a novel intelligent distributed network softwarization architecture for IoT is proposed and used to provide a new intelligent routing mechanism. Conducted simulations have proven the effectiveness of our proposal.

Keywords: Internet of Things (IoT), Network Softwarization, Software Defined Networking (SDN), Network Function Virtualization (NFV), Machine Learning (ML).

Heudiasyc UMR UTC/CNRS 7253

Université de Technologie de Compiègne (UTC)

Avenue de Landshut, CS 60319

60203 COMPIEGNE CEDEX

+33 (0)3 44 23 46 45

www.hds.utc.fr